

Ratio, Proportion and Partnership

The comparison between two quantities in terms of magnitude is called **ratio**.

For example, Mohit has 5 pens and Amit has 3 pens. It means the ratio of number of pens between Mohit and Amit is 5 is to 3. It can be expressed as 5 : 3.

So the ratio of any two quantities is expressed as $\frac{a}{b}$ or $a : b$. The numerator 'a' is called the antecedent and denominator 'b' is called as consequent.

Rule of Ratio: The comparison of two quantities is meaningless if they are not of the same kind or in the same units (of length, volume or currency etc.) We do not compare 5 girls and 7 toys or 15 kilometers and 3 cows. Therefore, to find the ratio of two quantities (of the same kind), it is necessary to explain them in same units.

Properties of Ratio:

1. The nature of ratio does not change when the numerator and denominator both are multiplied by same quantities.

$$\text{i.e., } \frac{a}{b} = \frac{ka}{kb} = \frac{la}{lb} \text{ etc.} \Rightarrow \text{e.g., } \frac{3}{4} = \frac{6}{8} = \frac{9}{12} \dots \text{etc have the same ratio.}$$

2. The value of a ratio does not change when the numerator and denominator both are divided by same quantities.

$$\text{i.e., } \frac{a}{b} = \frac{a/k}{b/k} = \frac{a/1}{b/1} \text{ etc.,} \Rightarrow \text{e.g., } \frac{3}{4} = \frac{3/2}{4/2} = \frac{3/3}{4/3} \dots \text{etc are in same ratio}$$

3. The ratio of two fractions can be expressed in ratio of integers.

$$\text{e.g., } \frac{3/4}{5/4} = \frac{3}{4} \times \frac{4}{5} = \frac{3}{5} \Rightarrow \frac{a/b}{c/d} = \frac{a}{b} \times \frac{d}{c}$$

4. When two or more than two ratio are multiplied with each other, then it is called compounded ratio e.g.,

$$\frac{2}{3} \times \frac{4}{5} \times \frac{6}{7} = \frac{16}{35} \text{ is compounded ratio of } \frac{2}{3}, \frac{4}{5}, \frac{6}{7}$$

$$\text{So, } \frac{a}{b} \times \frac{c}{d} \times \frac{e}{f} \dots = \frac{k}{m} \text{ (Compound ratio)}$$

5. When the ratio is compounded with itself, it is called as duplicate, triplicate ratio etc.

$$\frac{a}{b} \times \frac{a}{b} = \frac{a^2}{b^2} = \left(\frac{a}{b}\right)^2 \text{ is called duplicate ratio of } \frac{a}{b}$$

$$\text{and } \frac{a}{b} \times \frac{a}{b} \times \frac{a}{b} = \left(\frac{a}{b}\right)^3 \text{ is called triplicate ratio of } \frac{a}{b}$$

$$\text{similarly } \sqrt{\left(\frac{a}{b}\right)} = \left(\frac{a}{b}\right)^{\frac{1}{2}} \text{ is called as sub-duplicate ratio and}$$

$$\sqrt[3]{\left(\frac{a}{b}\right)} = \left(\frac{a}{b}\right)^{\frac{1}{3}} \text{ is called as sub-triplicate ratio of } \frac{a}{b}.$$

If four quantities a, b, c and d form a proportion, many other proportions may be deduced by the properties of fraction. The results of these operation are very useful. These operations are

1. Invertendo: if $\frac{a}{b} = \frac{c}{d}$ then $\frac{b}{a} = \frac{d}{c}$
2. Alternando: if $\frac{a}{b} = \frac{c}{d}$ then $\frac{a}{c} = \frac{b}{d}$
3. Componendo: if $\frac{a}{b} = \frac{c}{d}$, then $\left(\frac{a+b}{b}\right) = \left(\frac{c+d}{d}\right)$
4. Dividendo: if $\frac{a}{b} = \frac{c}{d}$, then $\left(\frac{a-b}{b}\right) = \left(\frac{c-d}{d}\right)$
5. Componendo and Dividendo: if $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Concept 1: If two numbers are in the ratio of $a : b$ and the sum of these number is x , then these numbers will

be $\frac{ax}{a+b}$ and $\frac{bx}{a+b}$ respectively.

Example : Two numbers are in the ratio of $4 : 5$. If sum of these two number is 810, find the numbers?

Sol. Ratio of two number = $4 : 5$

Sum = 810

$$\text{First number} = \frac{ax}{a+b} = \frac{4 \times 810}{4+5} = 360, \quad \text{Second number} = \frac{bx}{a+b} = \frac{5 \times 810}{4+5} = 450$$

Example: $a : b = 3 : 4$ and $b : c = 2 : 5$ Find $a : b : c$?

Sol.

$$\begin{array}{l} a : b = 3 : 4 \\ b : c = 2 : 5 \end{array} \xrightarrow{\text{LCM of 4 and 5 is 20}} \begin{array}{l} a : b = 15 : 20 \\ b : c = 8 : 20 \end{array}$$

$$a : b : c = 15 : 20 : 20$$

$$= 3 : 4 : 4$$

Example: $a : b = 1 : 2$, $b : c = 3 : 2$, and $c : d = 1 : 3$. Find $a : b : c : d$?

Sol.

$$\begin{array}{cccc} a & b & c & d \\ 1 & 2 & \rightarrow 2 & \rightarrow 2 \\ 3 & \leftarrow 3 & 2 & \rightarrow 2 \\ 1 & \leftarrow 1 & \leftarrow 1 & 3 \end{array}$$

$$a : b : c : d = 1 \times 3 \times 1 : 2 \times 3 \times 1 : 2 \times 2 \times 1 : 2 \times 2 \times 3$$

$$= 3 : 6 : 4 : 12$$

Partnership

Concept 1: If a group of n persons invested different amount for different period then their profit is in the ratio s $At_1 : Bt_2 : Ct_3 : Dt_4 : \dots : Xt_n$

Here first person invested amount A for t_1 period, second persons invested amount B for t_2 period and so on.

Example: A starts a business with Rs 2,000, B joins him after 3 months with Rs 4,000. C puts a sum of Rs 10,000 in the business for 2 months only. At the end of the year the business gave a profit of Rs 5600. How should the profit be divided among them?

Sol: Ratio of their profits ($A's : B's : C's$) = $2 \times 12 : 4 \times 9 : 10 \times 2 = 6 : 9 : 5$

Now, $6 + 9 + 5 = 20$

$$\text{Then A's share} = \frac{5600}{20} \times 6 = \text{Rs } 1680, \quad \text{B's share} = \frac{5600}{20} \times 9 = \text{Rs } 2520, \quad \text{C's share} = \frac{5600}{20} \times 5 = \text{Rs } 1400$$

Concept 2: If investments are in the ratio of $a : b : c$ and the timing of their investments in the ratio of $x : y : z$ then the ratio of their profits are in the ratio of $ax : by : cz$.

Example : A, B and C invested capital in the ratio $2 : 3 : 5$, the timing of their investments being in the ratio $4 : 5 : 6$. In what ratio would their profit be distributed?

Sol. We should know that if the duration for their investments be in the ratio $x : y : z$, and investment is in the ratio $a : b : c$ then the profit would be distributed in the ratio $ax : by : cz$.

Thus, following the same rule, the required ratio = $2 \times 4 : 3 \times 5 : 5 \times 6 = 8 : 15 : 30$

Concept 3: If investments are in the ratio $a : b : c$ and profits in the ratio $p : q : r$, then the ratio of time is $\frac{p}{a} : \frac{q}{b} : \frac{r}{c}$.

Example : A, B and C invested capital in the ratio 5 : 6 : 8. At the end of the business term, they received the profits in the ratio 5 : 3 : 12. Find the ratio of time for which they contributed their capital?

Sol: Using the above formula, we have the required ratio = $\frac{5}{5} : \frac{3}{6} : \frac{12}{8} = 1 : \frac{1}{2} : \frac{3}{2} = 2 : 1 : 3$.

Types of Questions

1. A sum of Rs 9000 is to be distributed among A, B and C in the ratio 4 : 5 : 6. What will be the difference between A's and C's shares?

Sol. Total amount = Rs 9000

A's share = $4x \Rightarrow$ B's share = $5x$

and C's share = $6x$

Then, $4x + 5x + 6x = 9000 \Rightarrow 15x = 9000; \therefore x = 600$

Now, A's share = $4 \times 600 =$ Rs 2400 and

C's share = $6 \times 600 =$ Rs 3600

Difference between A's and C's share

= Rs $(3600 - 2400) =$ Rs 1200

2. Rs 6400 are divided among three workers in the ratio

$\frac{3}{5} : 2 : \frac{5}{3}$. The share (in Rs) of the second worker is

Sol. Total amount = 6400

Let first worker's share = $\frac{3}{5}x$

Second worker's share = $2x$

Third worker's share = $\frac{5}{3}x$

Then, $\frac{3}{5}x + 2x + \frac{5}{3}x = 6400$

$$\Rightarrow \frac{9x + 30x + 25x}{15} = 6400$$

$$\Rightarrow 64x = 6400 \times 15 = 15000$$

$$\therefore \text{Second worker's share} = 1500 \times 2 = 3000$$

3. A boy, after giving away 80% of his pocket money to one companion and 6% of the remainder to another, has Rs 47 left with him. How much pocket money did the boy have in the beginning?

Sol. Let the boy had Rs x .

Money given to first companion = 80% of $x = \frac{4x}{5}$

Remaining money = $x - \frac{4x}{5} = \frac{x}{5}$

Money given to the another companion

$$= \frac{6}{100} \times \frac{x}{5} = \frac{3x}{250}$$

$$\text{Remaining money} = \frac{x}{5} - \frac{3x}{250} = \frac{50x - 3x}{250} = \frac{47x}{250}$$

$$\therefore \frac{47x}{250} = 47 \Rightarrow x = \frac{47 \times 250}{47} = \text{Rs } 250$$

4. Rs 180 contained in a box consists of Rs 1,50 paise and 25 paise coins in the ratio 2 : 3 : 4. What is the number of 50 paise coins?

Sol. Ratio of the values of the coins

$$= 2 : \frac{3}{2} : \frac{4}{4} = 2 : \frac{3}{2} : 1 = 4 : 3 : 2$$

Sum of the ratios = $4 + 3 + 2 = 9$

$$\therefore \text{Values of 50 paise coins} = \frac{3}{9} \times 180 = \text{Rs } 60$$

Numbers of 50 paise coins = 120

5. A, B and C enter into a partnership with shares in the

ratio $\frac{7}{2} : \frac{4}{3} : \frac{6}{5}$. After 4 months, A increase his share by

50%. If the total profit at the end of the year was Rs 43200. Then, the B's share in the profit is:

Sol. Ratio of initial shares of A, B and C in the partnership

$$A : B : C = \frac{7}{2} : \frac{4}{3} : \frac{6}{5} = \frac{7 \times 15}{2 \times 15} : \frac{4 \times 10}{3 \times 10} : \frac{6 \times 6}{5 \times 6}$$

$$= \frac{105}{30} : \frac{40}{30} : \frac{36}{30}$$

(LCM of 2, 3, 5 = 30) = 105 : 40 : 36

Let the respective shares of A, B and C be Rs 105x, Rs 40x and Rs 36x

New shares of A, B and C in the partnership

$$A = \text{Rs } 105x \text{ for 4 months} + 105x \times \frac{150}{100} \text{ for 8 months}$$

$$= (105x \times 4) + \left(105x \times \frac{3}{2} \times 8\right) = 420x + 1260x = 1680x$$

$$B = 40x \text{ for 12 months} = 40x \times 12 = 480x$$

$$C = 36x \text{ for 12 months} = 36x \times 12 = 432x$$

$$A : B : C = 1680 : 480 : 432 = 35 : 10 : 9$$

It is a type of simple partnership, so the profit or loss of the business is distributed among the investors in the ratio of their invested money.

∴ B's share in profit

$$= \frac{\text{B's investment}}{(\text{A} + \text{B} + \text{C})'s \text{ investment}} \times \text{Total profit}$$

$$= \frac{10}{35+10+9} \times 43200 = \frac{10}{54} \times 43200 = 8000$$

Hence, B's share in the profit = Rs 8000.

7. In a business partnership among A, B, C and D, the profit is shared as follows

$$\frac{\text{A's share}}{\text{B's share}} = \frac{\text{B's share}}{\text{C's share}} = \frac{\text{C's share}}{\text{D's share}} = \frac{1}{3}$$

If the total profit is Rs 400000, the share of C is:

- Sol.** Given, A : B = 1 : 3

$$B : C = 1 : 3 = 3 : 9$$

$$C : D = 1 : 3 = 9 : 27$$

Now, by joining the above three ratios, we get

$$A : B : C : D = 1 : 3 : 9 : 27$$

$$\text{Sum of the ratios} = 1 + 3 + 9 + 27 = 40$$

$$\therefore \text{C's share in profit} = \frac{9}{40} \times 400000 = \text{Rs } 90000$$

8. A started a business with a capital of Rs 100000. 1 yr later, B joined him with a capital of Rs 200000. At the end of 3 yr from the start of the business, the profit earned was Rs 84000. The share of B in the profit exceeded the share of A by

- Sol.** Ratio of equivalent capitals of A and B
= 100000 × 36 : 200000 × 24 = 36 : 48 = 3 : 4

$$\text{Profit gained by A} = \frac{3}{7} \times 84000 = \text{Rs. } 36000$$

$$\text{Profit gained by B} = \frac{4}{7} \times 84000 = \text{Rs. } 48000$$

$$\text{Required difference} = 48000 - 36000 = \text{Rs. } 12000$$

9. The ratio of income of A and B is 3 : 4. If the ratio of expenditure of both is 2 : 3 and each save Rs 200, find the income of both A and B?

- Sol.** Income - Saving = Expenditure

According to question

Let the income of A and B be 3x and 4x respectively

$$\frac{3x - 200}{4x - 200} = \frac{2}{3}, \quad x = 200$$

$$\text{Income of A} = 3x = 3 \times 200 = 600$$

$$\text{Income of B} = 4x = 4 \times 200 = 800$$

10. Rs 7800 distributed among A, B, and C. The share of

A is $\frac{3}{4}$ th of the share of B, and share of B is $\frac{2}{3}$ th of the share of C. Then, find the difference between share of B and C?

- Sol.** $A = \frac{3}{4} B \Rightarrow A : B = 3 : 4$ $B = \frac{2}{3} C \Rightarrow B : C = 2 : 3$

$$A : B : C = 6 : 8 : 12 = 3 : 4 : 6$$

$$\text{Share of B} = \frac{4}{13} \times 7800 = 2400$$

$$\text{Share of C} = \frac{6}{13} \times 7800 = 3600$$

$$\text{Difference of share of B and C} = \text{Rs. } 1200$$

Foundation

Questions

1. One year ago the ratio of Ramu and Somu age was 6 : 7 respectively. Four years hence their ratio would become 7 : 8. How old is Somu?
(a) 24 years (b) 30 years
(c) 32 years (d) 36 years
2. If 33% of A is equal to 55% of B then find the ratio of A and B?
(a) 3 : 5 (b) 5 : 3
(c) 3 : 4 (d) 5 : 4
3. What is the value of 68% of two fifth of 550?
(a) 149.6 (b) 150
(c) 154 (d) 158
4. If 24 is deducted from 45% of a number result becomes 48. What will be $\frac{3}{8}$ part of that number?
(a) Rs. 500 (b) Rs. 1500
(c) Rs. 2000 (d) Rs. 2400
5. 30% of a number is 190.8. What will be 175% of that number?
(a) 1113 (b) 1115
(c) 1502 (d) 1002
6. What will be 32% of $\frac{3}{8}$ th of 1000?
(a) 115 (b) 125
(c) 120 (d) 129
7. Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is:
(a) 2 : 5 (b) 3 : 5
(c) 4 : 5 (d) 6 : 7
8. A sum of money is to be distributed among A, B, C, D in the proportion of 5 : 2 : 4 : 3. If C gets Rs. 1000 more than D, what is B's share?
(a) Rs. 500 (b) Rs. 1500
(c) Rs. 2000 (d) Rs. 2400

9. If $0.75 : x :: 5 : 8$, then x is equal to:
 (a) 1.12 (b) 1.2
 (c) 1.25 (d) 1.30
10. The sum of three numbers is 98. If the ratio of the first to second is $2 : 3$ and that of the second to the third is $5 : 8$, Then the second number is:
 (a) 20 (b) 30
 (c) 48 (d) 58
11. If Rs. 872 is divided into three parts, proportional to $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$, then the first part is:
 (a) Rs. 182 (b) Rs. 190
 (c) Rs. 227.46 (d) Rs. 204
12. The fourth proportional to 5, 8, 15 is:
 (a) 18 (b) 24
 (c) 19 (d) 20
13. Two numbers are in the ratio $3 : 5$. If 9 is subtracted from each, then new numbers are in the ratio $12 : 23$. The smaller number is :
 (a) 27 (b) 33
 (c) 49 (d) 55
14. If a number is reduced by 40% it become two third of another number. What is ratio of the second number to first number?
 (a) $6 : 10$ (b) $9 : 8$
 (c) $8 : 9$ (d) $9 : 10$
15. $\frac{5a+3b}{2a-3b} = \frac{23}{5}$ then the value of $a : b$ is:
 (a) $1 : 2$ (b) $1 : 3$
 (c) $4 : 1$ (d) $2 : 3$
16. If $P : Q = 8 : 15$ and $Q : R = 3 : 2$, then find $P : Q : R$?
 (a) $8 : 15 : 7$ (b) $7 : 15 : 8$
 (c) $8 : 15 : 10$ (d) $10 : 15 : 8$
17. If $P : Q = 8 : 15$, $Q : R = 5 : 8$ and $R : S = 4 : 5$, then $P : S$ is equal to:
 (a) $4 : 15$ (b) $2 : 15$
 (c) $3 : 19$ (d) $7 : 15$
18. Find the 4th proportional to 4, 16 and 7?
 (a) 28 (b) 29
 (c) 22 (d) 25
19. Find the mean proportional between 9 and 64?
 (a) 25 (b) 24
 (c) 27 (d) 35
20. What will be the duplicate ratio of $2 : 7$?
 (a) $4 : 49$ (b) $49 : 4$
 (c) $4 : 14$ (d) $8 : 343$
21. Find the sub-duplicate ratio of $81 : 64$?
 (a) $8 : 9$ (b) $4 : 9$
 (c) $9 : 8$ (d) $7 : 8$
22. Find the triplicate ratio of $7 : 5$?
 (a) $125 : 343$ (b) $343 : 125$
 (c) $344 : 125$ (d) $343 : 126$
23. What will be the inverse ratio of $17 : 19$?
 (a) $19 : 17$ (b) $18 : 17$
 (c) $17 : 18$ (d) $19 : 5$
24. Find the compound ratio of $2 : 7$, $5 : 3$ and $4 : 7$?
 (a) $147 : 40$ (b) $40 : 147$
 (c) $147 : 30$ (d) $30 : 147$
25. If $A : B = 3 : 4$ and $B : C = 8 : 9$, then find the value of $A : B : C$?
 (a) $3 : 4 : 5$ (b) $1 : 2 : 3$
 (c) $7 : 12 : 17$ (d) $6 : 8 : 9$
26. If $a : b = 3 : 5$ and $b : c = 4 : 7$, then $a : c$ is equal to:
 (a) $11 : 35$ (b) $35 : 11$
 (c) $35 : 12$ (d) $12 : 35$
27. If $P : Q : R = 2 : 3 : 4$, then find $\frac{P}{Q} : \frac{Q}{R} : \frac{R}{P}$?
 (a) $8 : 9 : 24$ (b) $9 : 8 : 24$
 (c) $24 : 8 : 9$ (d) $8 : 24 : 9$
28. If $\frac{a}{3} = \frac{b}{8}$, then $(a + 3) : (b + 8)$ is equal to:
 (a) $3 : 8$ (b) $8 : 3$
 (c) $5 : 8$ (d) $3 : 5$
29. The ratio of $4^3 : 2^5$ is same is:
 (a) $2 : 1$ (b) $4 : 1$
 (c) $7 : 5$ (d) $7 : 10$
30. If $2 : x :: 5 : 7$, then x is equal to:
 (a) 1.12 (b) 2.80
 (c) 1.25 (d) 1.30
31. The salaries of A, B, C are in the ratio $2 : 3 : 5$. If the increments of 15%, 10% and 20% are allowed respectively in their salaries, then what will be the new ratio of their salaries?
 (a) $3 : 3 : 10$ (b) $10 : 11 : 20$
 (c) $23 : 33 : 60$ (d) $25 : 27 : 29$
32. If Rs. 782 be divided into three parts, proportional to $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$, then the first part is:
 (a) Rs. 182 (b) Rs. 190
 (c) Rs. 196 (d) Rs. 204
33. Two numbers are in the ratio $1 : 2$. If 7 is added to both, they changes to $3 : 5$. The greatest number is:
 (a) 24 (b) 26
 (c) 28 (d) 32
34. The ratio of three numbers is $3 : 4 : 5$ and the sum of their squares is 1250. The sum of the numbers is :
 (a) 30 (b) 50
 (c) 60 (d) 90

35. Sachin is younger than Rahul by 4 years. If their ages are in the respective ratio of 7 : 9, how old is Sachin?
 (a) 16 years (b) 18 years
 (c) 14 years (d) 15 years
36. At present, the ratio between the ages of Arun and Deepak is 4 : 3. After 6 years. Arun's age will be 26 years. What is the age of Deepak at present?
 (a) 12 years (b) 15 years
 (c) $19\frac{1}{2}$ years (d) 21 years
37. Present ages of X and Y are in the ratio 5 : 6 respectively. Seven year hence this ratio will become 6 : 7 respectively. What is X's present age in years?
 (a) 35 (b) 42
 (c) 49 (d) Cannot be determined
38. Present ages of Sameer and Anand are in the ratio of 5 : 4 respectively. Three years hence, the ratio of their ages will become 11 : 9 respectively. What is Anand's present age in years?
 (a) 24 (b) 27
 (c) 40 (d) Cannot be determined
39. The total of the ages of Jayant, Prem and Saransh is 93 years. Ten years ago, the ratio of their ages was 2 : 3 : 4. What is the present age of Saransh?
 (a) 24 years (b) 32 years
 (c) 34 years (d) 38 years
40. Hitesh is 40 years old and Ronnie is 60 years old. How many years ago was the ratio of their ages 3 : 5?
 (a) 5 years (b) 10 years
 (c) 20 years (d) 37 years
41. Aditya and Sanjay started a business investing 45000 and 30000 respectively. What will be the ratio between their profit?
 (a) 2 : 3 (b) 3 : 2
 (c) 4 : 9 (d) Cant be determined
42. Mahesh and Kishore started a business with investing Rs. 60,000 and Rs. 50,000 respectively. If after 1 year they earned Rs. 33,000 profit, what will be kishore share?
 (a) 15,000 (b) 30,000
 (c) 35,000 (d) 20,000
43. Anil and Nikhil started a business with investing Rs. 3300 and Rs. 5000 respectively. If they earned profit of 1660 then what will be Anil's share?
 (a) 640 (b) 650
 (c) 655 (d) 660
44. Rahul started a business by investing Rs. 45,000. 6 months later Sanjay joined him with Rs. 30,000. In what ratio should the earned profit be distributed at the end of the year?
 (a) 1 : 3 (b) 3 : 1
 (c) 2 : 3 (d) 3 : 2
45. Anil, Nikhil and Saurabh started a business in partnership by investing in the ratio of 6 : 5 : 8 respectively. At the end of the year they earned a profit of Rs. 83,600. What will be Saurabh's share?
 (a) Rs. 30,200 (b) Rs. 35,200
 (c) Rs. 32,500 (d) Rs. 34,000
46. Ram, Karan and Rohan invested capital in the ratio of 2 : 3 : 4 for time period of 6 : 4 : 3. Find the ratio of profit distributed?
 (a) 12 : 13 : 14 (b) 13 : 12 : 14
 (c) 1 : 1 : 1 (d) 14 : 12 : 13
47. Ravi and Kavi start a business by investing Rs. 8000 and Rs. 72000, respectively. Find the ratio of their profits at the end of year.
 (a) 2 : 9 (b) 5 : 9
 (c) 7 : 9 (d) 1 : 9
48. Srikant and Vividh started a business investing amounts of Rs. 185000 and Rs. 225000, respectively. If Vividh's share in the profit earned by them is Rs. 9000, what is the total profit earned by them together?
 (a) Rs. 17400 (b) Rs. 16400
 (c) Rs. 16800 (d) Rs. 17800
49. Rajan and Sajan started a business initially with Rs. 14200 and Rs. 15600, respectively. If total profits at the end of year is Rs. 74500, what is the Rajan's share in the profit?
 (a) Rs. 39000 (b) Rs. 39600
 (c) Rs. 35000 (d) Rs. 35500
50. A starts a business with Rs. 4000 and B joins him after 3 months with Rs. 16000. Find the ratio of their profits at the end of year.
 (a) 1 : 3 (b) 2 : 33
 (c) 1 : 9 (d) 1 : 7

Moderate

1. The difference between a two-digit number and the number obtained by interchanging the digits is 36. What is the difference between the sum and the difference of the digits of the number if the ratio between the digits of the number is 1 : 2 ?
 (a) 4 (b) 8
 (c) 16 (d) 20
2. Seats for Mathematics, Physics and Biology in a school are in the ratio 5 : 7 : 8. There is a proposal to increase these seats by 40%, 50% and 75% respectively. What will be the ratio of increased seats?
 (a) 2 : 3 : 4 (b) 6 : 7 : 8
 (c) 6 : 8 : 9 (d) 4 : 8 : 9
3. In a mixture 60 litres, the ratio of milk and water is 2 : 1. What quantity of water should be added so that the ratio becomes 1 : 3?
 (a) 80 L (b) 100 L
 (c) 120 L (d) 60 L
4. The ratio of the number of boys and girls in a college is 7 : 8. If the percentage increase in the number of boys and girls be 20% and 10% respectively, what will be the new ratio?
 (a) 8 : 9 (b) 17 : 18
 (c) 21 : 22 (d) Can't be determined
5. Salaries of Ravi and Sumit are in the ratio 2 : 3. If the salary of each is increased by Rs. 4000, the new ratio becomes 40 : 57. What is Sumit's salary?
 (a) Rs. 17,000 (b) Rs. 20,000
 (c) Rs. 25,500 (d) Rs. 34,000
6. The salaries of A, B and C are in the ratio 2 : 3 : 5. If the increments of 15%, 10% and 20% are allowed respectively in their salaries, then what will be new ratio of their salaries?
 (a) 3 : 3 : 10 (b) 10 : 11 : 20
 (c) 23 : 33 : 60 (d) 3 : 4 : 5
7. In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 2 : 3 : 4. If there is Rs. 50 in all, how many 5 p coins are there?
 (a) 50 (b) 100
 (c) 50 (d) 200
8. A sum of money is divided among C, A and B in ratio of 4 : 5 : 6 and another sum of money is divided between M and N equally if B got 2000 more than M then how much C get?
 (a) Rs. 1000 (b) Rs. 500
 (c) Rs. 4000 (d) Can't be determined
9. Three number A, B and C are in ratio of 12 : 15 : 25. If sum of these numbers be 364 find the ratio between difference of B and A and the difference of C and B?
 (a) 3 : 2 (b) 3 : 10
 (c) 3 : 5 (d) 4 : 2
10. $\frac{625}{x} = \frac{x}{1156}$ then what will be the value of x
 (a) 800 (b) 850
 (c) 900 (d) 950
11. A mixture contains alcohol and water in the ratio 4 : 3. If 5 liters of water is added to the mixture, the ratio becomes 4 : 5. Find the quantity of alcohol in the given mixture.
 (a) 5 Ltr. (b) 7.5 Ltr.
 (c) 10 Ltr. (d) 12 Ltr.
12. In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there are Rs. 30 in all, how many 5 p coins are there?
 (a) 50 (b) 100
 (c) 150 (d) 200
13. If Rs. 510 be divided among A, B, C in such a way that A gets $\frac{2}{3}$ of what B gets and B gets $\frac{1}{4}$ of what C gets, then their shares are respectively:
 (a) Rs. 120, Rs. 240, Rs. 150
 (b) Rs. 60, Rs. 90, Rs. 360
 (c) Rs. 150, Rs. 300, Rs. 60
 (d) Rs. 70, Rs. 90, Rs. 350
14. Rs. 366 are divided among A, B and C so that A may get $\frac{1}{2}$ as much as B and C together, B may get $\frac{2}{3}$ as much as A and C together, then the share of A is :
 (a) Rs. 122 (b) Rs. 129.60
 (c) Rs. 146. 60 (d) Rs. 183
15. Ratio of the earnings of A and B is 4 : 7. If the earnings A increase by 50% and those of B decrease by 25%, the new ratio of their earnings becomes 8 : 7. What are A's earnings?
 (a) Rs. 21,000 (b) Rs. 26,000
 (c) Rs. 28,000 (d) Data inadequate
16. An amount of Rs. 735 was divided between A, B and C. If each of them had received Rs. 25 less, their shares would have been in the ratio of 1 : 3 : 2. The money received by C was :
 (a) Rs. 195 (b) Rs. 200
 (c) Rs. 225 (d) Rs. 245
17. An amount of Rs. 2430 is divided among A, B and C such that if their shares be reduced by Rs. 5, Rs. 10 and Rs. 15 respectively, the remainders shall be in the ratio of 3 : 4 : 5. Then, B's share was:
 (a) Rs. 605 (b) Rs. 790
 (c) Rs. 800 (d) Rs. 810

18. Gold is 19 times as heavy as water and copper is 9 times as heavy as water. In what ratio should these be mixed to get an alloy 15 times as heavy as water ?
 (a) 1 : 1 (b) 2 : 3
 (c) 1 : 2 (d) 3 : 2
19. 15 litres of mixture contains 20% alcohol and the rest water. If 3 litres of water be mixed with it, the percentage of alcohol in the new mixture would be :
 (a) 15% (b) $16\frac{2}{3}\%$
 (c) 17% (d) $18\frac{1}{2}\%$
20. 85 L of a mixture contains milk and water in the ratio 27 : 7. How much more water is to be added to get a new mixture containing milk and water in the ratio 3:1 ?
 (a) 5 L (b) 6.5 L
 (c) 7.25 L (d) 8 L
21. The sides of a triangle are in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$ and its perimeter is 104 cm. The length of the longest side is :
 (a) 52 cm (b) 48 cm
 (c) 32 cm (d) 26 cm
22. The ratio of the number of boys and girls in a school is 3 : 2. If 20% of the boys and 25% of the girls are scholarship holders, what percentage of the students does not get the scholarship?
 (a) 56 (b) 70
 (c) 78 (d) 80
23. Three containers have volumes in the ratio 3 : 4 : 5. They are full of mixtures of milk and water. The mixtures contain milk and water in the ratio of 4 : 1, 3:1 and 5 : 2 respectively. The contents of all these three containers are poured into a fourth container. The ratio of milk and water in the fourth container is:
 (a) 4 : 1 (b) 151 : 48
 (c) 157 : 53 (d) 5 : 2
24. x varies inversely as square of y. Given that y = 2 for x = 1. The value of x for y = 6 will be equal to :
 (a) 3 (b) 9
 (c) $\frac{1}{3}$ (d) $\frac{1}{9}$
25. The electricity bill of a certain establishment is partly fixed and partly varies as the number of units of electricity consumed. When in a certain month 500 units are consumed, the bill is Rs. 1680. In another month 600 units are consumed and the bill is Rs. 1980. In yet another month 700 units are consumed. The bill for that month would be:
 (a) Rs. 2280 (b) Rs. 1680
 (c) Rs. 1840 (d) Rs. 1950
26. The ratio of the incomes of A and B is 5 : 4 and the ratio of their expenditures is 3 : 2. If at the end of the year, each saves Rs. 1600, then the income of A is :
 (a) Rs. 3400 (b) Rs. 3600
 (c) Rs. 4000 (d) Rs. 4400
27. A and B are two alloys of gold and copper prepared by mixing metals in the ratio 7 : 2 and 7 : 11 respectively. If equal quantities of the alloys are melted to form a third alloy C, the ratio of gold and copper in C will be :
 (a) 5 : 7 (b) 5 : 9
 (c) 7 : 5 (d) 9 : 5
28. The ratio between the present ages of P and Q is 5 : 7 respectively. If the difference between Q's present age and P's age after 6 years is 2, what is the total of P's and Q's present ages?
 (a) 48 years (b) 52 years
 (c) 56 years (d) Cannot be determined
29. The ratio of the father's age to his son's age is 7 : 3. The product of their ages is 756. The ratio of their ages after 6 years will be :
 (a) 5 : 2 (b) 2 : 1
 (c) 11 : 7 (d) 13 : 9
30. The present ages of three persons are in proportions 4 : 7 : 9. Eight years ago, the sum of their ages was 56. Find their present ages (in years).
 (a) 8, 20, 28 (b) 16, 28, 36
 (c) 20, 35, 45 (d) 13, 16, 25
31. The ratio of the ages of a man and his wife is 4 : 3. After 4 years, this ratio will be 9 : 7. If at the time of marriage, the ratio was 5 : 3, then how many years ago were they married?
 (a) 8 years (b) 10 years
 (c) 12 years (d) 15 years
32. The ratio between the present ages of A and B is 5 : 3 respectively. The ratio between A's age 4 years ago and B's age 4 years hence is 1 : 1. What is the ratio between A's age 4 years hence and B's age 4 years ago ?
 (a) 1 : 3 (b) 2 : 1
 (c) 3 : 1 (d) 4 : 1
33. A person's present age is two-fifth of the age of his mother. After 8 years, he will be one-half of the age of his mother. How old is the mother at present ?
 (a) 32 years (b) 36 years
 (c) 40 years (d) 48 years
34. Four years ago, the father's age was three times the age of his son. The total of the ages of the father and the son after four years, will be 64 years. What is the father's age at present?
 (a) 32 years (b) 36 years
 (c) 44 years (d) 40 years

35. The present ages of Reena and Usha are 24 and 36 yr, respectively. What was the ratio between the ages of Usha and Reena respectively 8 yr ago?
 (a) 7 : 4 (b) 4 : 7
 (c) 11 : 8 (d) 8 : 11
36. At present Anil is 1.5 times of Purvi's age. 8 yr hence, the respective ratio between Anil and Purvi's ages will be 25 : 18. What is Purvi's present age?
 (a) 50 yr (b) 28 yr
 (c) 42 yr (d) 36 yr
37. The difference between the ages of Meena and Seema is 3 yr and the ratio between their ages is 7 : 8. What is the sum of their ages?
 (a) 43 yr (b) 41 yr
 (c) 45 yr (d) 48 yr
38. The ratio of the present age of Manoj to that of Wasim is 3 : 11. Wasim is 12 yr younger than Rehana. Rehana's age after 7 yr will be 85 yr. What is the present age of Manoj's father, who is 25 yr older than Manoj?
 (a) 43 yr (b) 67 yr
 (c) 45 yr (d) 69 yr
39. The ratio between the present ages of Indira and Lizzy is 3 : 8, respectively. After 8 yr, Indira's age will be 20 yr. What was Lizzy's age 5 yr ago?
 (a) 37 yr (b) 27 yr
 (c) 28 yr (d) 38 yr
40. At present, Kavita is twice Sarita's age. 8 yr hence, the respective ratio between Kavita's and Sarita's ages will be 22 : 13. What is Kavita's present age?
 (a) 26 yr (b) 18 yr
 (c) 42 yr (d) 36 yr
41. 10 yr before, the ratio of ages of A and B was 13 : 17. After 17 yr from now, the ratio of their ages will be 10 : 11. The present age B is
 (a) 23 yr (b) 40 yr
 (c) 27 yr (d) 44 yr
42. Ram, Rohan and Karn invested in business in the ratio of 3 : 2 : 5 respectively. If Ram earn 100% more profit than Rohan and Karn earns 40% more profit than Rohan. What will be the share of Rohan in profit?
 (a) 2350 (b) 3640
 (c) 1235 (d) Can't be Determined
43. Ram started a business by investing Rs. 45000. After 3 Months, Sanjay joined him with a capital of Rs. 60,000. After 6 Months, Aditya joined them with a capital of Rs. 90,000. At the end of the year they earned a profit of Rs. 16500. What is the Ram share in business?
 (a) 6600 (b) 5900
 (c) 5500 (d) 5000
44. Ram and Anil enter into partnership for a year. Ram invests Rs. 1200 and Anil invests Rs. 1500. After 3 month, Aditya enters with Contribution of Rs. 900. If Anil withdraws his Money after 6 Months. What will be the share of Aditya, if the profit is Rs. 2450?
 (a) Rs. 700 (b) Rs. 1820
 (c) Rs. 630 (d) Rs. 1120
45. A bag contains 10-paise and 25-paise coins in the ratio of 17 : 6. If the total money in the bag is Rs. 112, the number of 10-paise coins is
 (a) 35 (b) 210
 (c) 490 (d) 595
46. P & Q enter into a partnership. P puts in Rs. 50 and Q puts Rs. 45. At the end of the 4 months P withdraws half his capital and at the end of 5th month Q withdraws half of his capital. S then enter into with the capital of Rs. 70. At the end of 12 months they got profit of Rs. 1272.5 What was S share in the profit?
 (a) 460 (b) 480
 (c) 490 (d) Can't be determined
47. A and B enter into a partnership by investing Rs. 12000 and Rs. 16000 respectively. After 4 months B withdraws Rs. 6000 and A invest Rs. 5000 more. After 2 more months C joined with Rs. 20,000 capital. What will be A's share in profit if total profit of Rs. 30,100 is earned at the end of one year?
 (a) 12360.5 (b) 12365
 (c) 12362.5 (d) Can't be determined
48. P, Q and R invested their capital in the ratio of 8 : 6 : 5 at the end of the business they received the profit in the ratio of 1 : 3 : 5. Find the ratio of time for which they contributed their capital?
 (a) 1 : 4 : 8 (b) 4 : 1 : 8
 (c) 8 : 4 : 1 (d) Can't be determined
49. Aditya and Manish continued in a joint business for 36 months. Aditya contribute Rs. 300 for certain time and Manish invested Rs. 500 for remaining time. If out of total profit of Rs. 1020, Aditya gets Rs. 495, then for how long Aditya kept his money?
 (a) 16 months (b) 14 months
 (c) 8 months (d) 22 months
50. Aditya, Manish and Gaurav enter into partnership with shares in the ratio of 7/2 : 4/3 : 6/5. After 4 months, Aditya increase his share by 50%. If at the end of one year profit be Rs. 43200. Then what will be Manish share in profit?
 (a) 8000 (b) 7200
 (c) 28000 (d) 24400

Difficult

1. An alloy of copper and zinc is taken in the ratio 1 : 2, and another alloy of the same metal is taken in the ratio of 2 : 3. How many parts of the two alloys must be taken to obtain a new alloy consisting of copper and zinc that are in the ratio 3 : 5?
(a) 7 and 9 (b) 5 and 7
(c) 3 and 5 (d) 5 and 3
2. In two alloys, gold and silver are in the ratios of 4 : 1 and 1 : 3. After alloying together 10 kg. of the first alloy, 16 kg. of the second and several kilograms of pure gold, an alloy was obtained in which the ratio of gold to silver was 3 : 2. Find the weight of the new alloy.
(a) 12 kg. (b) 19 kg.
(c) 24 kg. (d) 35 kg.
3. Rs. 5625 are divided among A, B and C so that A received half as much as B and C together and B received one fourth as much as A and C together. Find the sum of A's and B's share together.
(a) Rs. 2500 (b) Rs. 3000
(c) Rs. 1500 (d) Rs. 4000
4. A man spends Rs. 2310 on an average during the first 8 month from the income. During next 4 months he spends Rs. 1800 on an average from his income but he had taken a loan of Rs. 1680 apart from his income to meet his both ends. If he spent all what he had then the ratio of average income and average expenditure?
(a) 50 : 53 (b) 100 : 103
(c) 100 : 109 (d) 100 : 107
5. In a family the ratio of expenses to the savings is 5 : 3. But his expenses is increased by 60% and income is increased by only 25% thus there is decrease of Rs. 3500 in the saving. Find the increased income of the family.
(a) Rs. 35000 (b) Rs. 28000
(c) Rs. 25000 (d) Rs. 18,500
6. Arti has coins of the denomination of Rs. 1, 50 paise and 25 paise in the ratio of 12 : 10 : 7. The total worth of the coins she has is Rs. 75. Find the number of 25 paise coins.
(a) 28 (b) 36
(c) 42 (d) 32
7. Two horses cost as much as 5 dogs, 6 dogs as much as 8 oxen, 10 oxen as much as 50 sheep, 14 sheep as much as 9 goats. If the price of one goat is Rs. 700, find the cost of one horse.
(a) Rs. 7500 (b) Rs. 6400
(c) Rs. 5200 (d) Rs. 4800
8. Rs. 2186 is distributed among A, B and C. If money given to them is decreased by Rs. 26, Rs. 28 and Rs. 32 respectively, then the ratio of their share become 9 : 13 : 8. What is the amount given to A ?
(a) Rs. 696 (b) Rs. 626
(c) Rs. 656 (d) Rs. 956
9. Sita and Gita invested the same capital in a business. At the year-end they share the profit in the ratio of 3 : 2. If Sita has invested her capital for the whole year, for how many months Gita has invested her capital?
(a) 8 months (b) 7 months
(c) 5 months (d) 9 months
10. A year ago, the cost of Maruti and Figo are in the ratio of 3 : 4. The ratio of present and past year costs of Maruti and Figo are 5 : 4 and 3 : 2. If the sum of their present cost is 7.8 lacs. Find the cost of Figo a year ago.
(a) 3.2 lac (b) 1.6 lac
(c) 3.5 lac (d) 4.0 lac
11. One year ago, the ratio between Ram's and Shyam's salaries was 3 : 5. The ratio of their individual salaries of last year and present year are 2 : 3 and 4 : 5 respectively. If their total salaries for the present year is Rs. 8600, find the present salary of Ram ?
(a) 3200 (b) 3600
(c) 4000 (d) 4400
12. Two candles of the same height are lighted at the same time. The first is consumed in 8 hours and the second in 6 hours. Assuming that each candle burns at a constant rate, in how many hours after being lighted, the ratio between the height of first and second candles becomes 2 : 1?
(a) 4 h (b) 5 h
(c) 4 h 30 m (d) 4 h 48m
13. The total of the daily wages of Ram, Shyam and Mohan is Rs. 450. If they, respectively spend 25%, 20% and 50% of their incomes then the ratio of their savings becomes 9 : 8 : 4. What is the income of Ram?
(a) 180 (b) 190
(c) 200 (d) 175
14. There is 1000 litres milk in a pot. 100 litres of milk is taken out and same amount of water is poured into it. Again 200 litres of mixture is taken out and same amount of water is poured into it and finally 400 litres of mixture is taken out and same amount of water is poured into it then what is the amount of milk in the resulting mixture?
(a) 432 (b) 436
(c) 425 (d) 438
15. On entire Earth the ratio between land and water is 1 : 2 and at northern hemisphere this ratio is 2 : 3. What will be the ratio of land with water at southern hemisphere?
(a) 11 : 4 (b) 4 : 11
(c) 15 : 4 (d) 4 : 15

16. The ratio of the incomes of A, B and C is 3 : 7 : 4 and the ratio of their expenditures is 4 : 3 : 5. If in the income of Rs. 2400, A saves Rs. 300 then the savings of B and C is respectively are
 (a) Rs. 4025 and Rs. 575
 (b) Rs. 1575 and 2625
 (c) Rs. 2750 and Rs. 1025
 (d) Rs. 3725 and Rs. 1525
17. The ratio of wine and water in a drum is 3 : 1. What part of the mixture is taken out and same amount of water is added to it so that the ratio of wine and water in the mixture become 1 : 1?
 (a) $\frac{1}{4}$ (b) $\frac{1}{3}$
 (c) $\frac{3}{4}$ (d) $\frac{2}{3}$
18. In a mixture of petrol and kerosene, petrol is only 99 litres. If this same quantity of petrol would be presented in another mixture of petrol and kerosene where total volume would be 198 litres less than the actual mixture then the concentration of petrol in the actual mixture would have been 13.33% less than that of the new mixture. What is the concentration of petrol in actual mixture?
 (a) 20% (b) 16.66%
 (c) 26.66% (d) 8.33%
19. The ratio of age between A and B is 6 : 5 and the age of each C and D is $\frac{9}{10}$ times that of B. Age of F is less than A but greater than B. The ratio of ages between B and E is 2 : 3. Also age of A is 3 years less than E. What is the ratio of ages of A and F if all the ages are in integers?
 (a) 12 : 11 (b) 9 : 7
 (c) 24 : 19 (d) 12 : 13
20. Three vessels having volumes in the ratio of 2 : 3 : 5 are full of a mixture of water and milk. In the first vessel ratio of water and milk is 1 : 3, in second 2 : 3 and in third vessel, 2 : 5. If all the three vessels were poured out in a large container, what is the resulting ratio of milk and water?
 (a) 43 : 96 (b) 438 : 962
 (c) 348 : 962 (d) 481 : 219

Previous Year (Memory Based)

1. If $A : B = 5 : 7$ and $B : C = 6 : 11$, then $A : B : C$ is :
 (a) 55 : 77 : 66 (b) 30 : 42 : 77
 (c) 35 : 49 : 42 (d) None of these
2. If $A : B : C = 2 : 3 : 4$, then $\frac{A}{B} : \frac{B}{C} : \frac{C}{A}$ is equal to :
 (a) 4 : 9 : 16 (b) 8 : 9 : 12
 (c) 8 : 9 : 16 (d) 8 : 9 : 24
3. The ratio of $4^{3.5} : 2^5$ is same as :
 (a) 2 : 1 (b) 4 : 1
 (c) 7 : 5 (d) 7 : 10
4. If $(x : y) = 2 : 1$, then $(x^2 - y^2) : (x^2 + y^2)$ is :
 (a) 3 : 5 (b) 5 : 3
 (c) 1 : 3 (d) 3 : 1
5. If $(4x^2 - 3y^2) : (2x^2 + 5y^2) = 12 : 19$, then $(x : y)$ is :
 (a) 2 : 3 (b) 1 : 2
 (c) 3 : 2 (d) 2 : 1
6. If $(a + b) : (b + c) : (c + a) = 6 : 7 : 8$ and $(a + b + c) = 14$, then the value is c is :
 (a) 6 (b) 7
 (c) 8 (d) 14
7. If Rs. 782 be divided into three parts, proportional to $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$ then the first part is :
 (a) Rs 182 (b) Rs. 190
 (c) Rs. 196 (d) Rs. 204
8. Salaries of Ravi and Sumit are in the ratio 2 : 3. If the salary of each is increased by Rs. 4000, the new ratio becomes 40 : 57. What is Sumit's present salary?
 (a) Rs. 17,000 (b) Rs. 20,000
 (c) Rs. 25,500 (d) None of these
9. If Rs. 510 be divided among A, B, C in such a way that A gets $\frac{2}{3}$ of what B gets and B gets $\frac{1}{4}$ of what C gets, then their shares are respectively:
 (a) Rs. 120, Rs 240, Rs. 150
 (b) Rs. 60, Rs. 90, Rs. 360
 (c) Rs. 150, Rs. 300, Rs 60
 (d) None of these
10. The ratio of the number of boys and girls in a school is 3 : 2. If 20% of the boys and 25% of the girls are scholarship holders, what percentage of the students does not get the scholarship?
 (a) 56 (b) 70
 (c) 78 (d) 80
11. In a school, 10% of the boys are same in number as $\frac{1}{4}$ th of the girls. What is the ratio of the boys to girls in that school?
 (a) 3 : 2 (b) 5 : 2
 (c) 2 : 1 (d) 4 : 3

12. The electricity bill of a certain establishment is partly fixed and partly varies as the number of units of electricity consumed. When in a certain month 540 units are consumed, the bill is Rs. 1800. In another month 620 units are consumed and the bill is Rs. 2040. In yet another month 500 units are consumed. The bill for that month would be :
 (a) Rs. 1560 (b) Rs. 1680
 (c) Rs. 1840 (d) Rs. 1950
13. A sum of Rs. 53 is divided among A, B, C in such a way that A gets Rs. 7 more than what B gets and B gets Rs. 8 more than what C gets. The ratio of their shares is :
 (a) 16 : 9 : 18 (b) 25 : 18 : 10
 (c) 18 : 25 : 10 (d) 15 : 8 : 30
14. In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there are Rs. 30 in all, how many 5 p coins are there?
 (a) 50 (b) 100
 (c) 150 (d) 200
15. There is 34000 rupees in a bag in the denomination of Rs. 50, Rs. 100, Rs. 500 and Rs. 1000 currency notes. If the ratio of the number of these notes is 2 : 3 : 4 : 1 then find the number of notes of 500 denomination.
 (a) 20 (b) 40
 (c) 30 (d) 50
16. In a cricket tournament three batsman A, B and C scored a total of 1750 runs. C scored three-fourth runs of total runs scored by A and B together. While B scored $\frac{12}{23}$ part runs of total runs scored by A and C together. How many runs A batsman scored?
 (a) 400 (b) 500
 (c) 200 (d) 600
17. There are 68 employees in a office of class -1, class -2 and class -3 who are in the ratio of 5 : 10 : 2 respectively. Their total salary is Rs. 400000 which is in the ratio of 2 : 3 : 5 respectively. What is the salary of employee of class-3?
 (a) 40000 (b) 4000
 (c) 10000 (d) 6000
18. A person travelled a distance of 68 km by an auto-rickshaw, a car and a bullock-cart in the ratio of 5 : 10 : 2. The fares of these vehicles per kilometer are in the ratio of 2 : 3 : 5 then how much money he paid to car vehicle if he totally spend Rs. 400?
 (a) Rs. 240 (b) Rs. 150
 (c) Rs. 340 (d) Rs. 300
19. The price of diamond is directly proportional to the square of its weight. A diamond broke into 4 pieces in the ratio of 1 : 2 : 3 : 4. The broken pieces got sold in Rs. 2,10,000 less. What was the actual price of the diamond?
 (a) 3,00,000 (b) 5,00,000
 (c) 2,00,000 (d) 6,00,000
20. If $\frac{a+b}{5} = \frac{b+c}{6} = \frac{c+a}{7}$ then find the value of a : b : c and if $a + b + c = 18$ then find the value of b.
 (a) 4 (b) 2
 (c) 6 (d) 5
21. If $2a = 3b = 4c$ then find the value of a : b : c
 (a) 6 : 4 : 3 (b) 4 : 3 : 1
 (c) 2 : 4 : 3 (d) 2 : 6 : 4
22. There are 1 rupee, 50 paise and 25 paise coins in a bag. The ratio of number of 1 rupee and 50 paise coins is 5 : 6, while the ratio of number of 50 paise and 25 paise coins is 3 : 5. If there is total Rs. 262.50 in the bag then find the number of coins in 25 paise?
 (a) 250 coins (b) 540 coins
 (c) 220 coins (d) 650 coins
23. An amount of Rs. 11,700 was distributed among 10 males, 15 females and 5 children in such a way that share of 3 males is equal to the share of 4 females and the share of 6 children is equal to the share of 5 females. How much amount will each child get?
 (a) 300 (b) 600
 (c) 500 (d) 700
24. The ratio of the incomes of Ram and Shyam is 5 : 4 while the ratio of their expenditure is 10 : 7. If each of them saves Rs. 2500 then find the income of Shyam income.
 (a) Rs. 6000 (b) Rs. 4000
 (c) Rs. 5000 (d) Rs. 3000
25. A milkman A, B and C hire a pasture in Rs. 2670. A grazes 4 cows and 3 calves for 5 days, B grazes 5 cows and 4 calves for 3 days while C grazes 6 cows and 5 calves for only 2 days. If one cow grazes three times of a calf then. What will be the difference between the rent of A and C?
 (a) Rs. 435 (b) Rs. 250
 (c) Rs. 500 (d) Rs. 300
26. Om, Jai and Jagdish start a business together. Om invests Rs. 5,000 for whole year. Jai invests Rs. 3,000 initially but increases it to 4,000 after 4 months. Jagdish invests Rs. 7,000 initially, but takes Rs. 4,000 back after 8 month. If their total profit at the end of the year is Rs. 34,400 then what are Jagdish's dividend?
 (a) Rs. 13,600 (b) Rs. 10,000
 (c) Rs. 15,000 (d) Rs. 12,000
27. Ram, Krishna and Ganesh started a business by investing capital in the ratio 5 : 4 : 3. Three months later Ram took back $\frac{2}{5}$ of his capital; four months

- later Krishna took back $\frac{1}{4}$ part of his capital while Ganesh invested $\frac{1}{3}$ part of his initial capital again. After 9 months there was a total profit of Rs. 19,200 in the business. How much the maximum profit?
- (a) Rs. 6000 (b) Rs. 6600
(c) Rs. 5000 (d) Rs. 1200
28. A and B started a business after investing Rs. 12,000 and Rs. 14,400 respectively. At the end of the year B gets 12% of the total profit as management fees and the rest amount is distributed between both the partners. If total annual profit is Rs. 7,500, then find the difference between their profits.
(a) Rs. 1400 (b) Rs. 1033
(c) Rs. 1500 (d) Rs. 1405
29. In a joint family the capitals invested by Laxmi and Saraswati are in the ratio 7 : 5 and the ratio of their profits is 4 : 3. If Laxmi's capital was invested for 20 months then for how much time Saraswati's capital would have been invested ?
(a) 21 months (b) 40 months
(c) 30 months (d) 15 months
30. If $p : q = r : s = t : u = 2 : 3$ then find the value of $(mp + nr + ot) : (mq + ns + ou)$
(a) 1 : 3 (b) 1 : 2
(c) 2 : 3 (d) 3 : 2
31. If $\sqrt{2} : (1 + \sqrt{3}) :: \sqrt{6} : x$, then what is the value of x?
(a) $3 + \sqrt{3}$ (b) $3 - \sqrt{3}$
(c) $3\sqrt{3}$ (d) $2 + \sqrt{3}$
32. Vinay started a business investing Rs 50000. After 1 year, he invested another Rs 30000 and Aditya also joined him with a capital of Rs 70000. If the profit earned in 3 years from the starting of business was Rs 87500, find out the share of Aditya in the profit?
(a) Rs 37500 (b) Rs 32500
(c) Rs 38281 (d) None of these
33. Average score of Rahul, Manish and Suresh is 63. Rahul's score is 15 less than Ajay and 10 more than Manish. If Ajay scored 30 marks more than the average scores of Rahul, Manish and Suresh, what is the sum of Manish's and Suresh's scores?
(a) 120 (b) 111
(c) 117 (d) Cannot be determined
34. A and B enter into partnership with capitals in the ratio 5 : 6. At the end of 8 months A withdraws his capital. They received profits in the ratio 5 : 9. Then for how much time, B remained in business?
(a) 6 months (b) 8 months
(c) 10 months (d) 12 months
35. A, B and C started a business investing Rs 42000, Rs 30000 and Rs 28000, respectively. After 4 months, A withdrew Rs 12000, B withdrew Rs 6000 and C withdrew Rs 8000. If, after 10 months, a total profit of Rs 46420 is earned, what is the share of C?
(a) Rs 12580 (b) Rs 13160
(c) Rs 13020 (d) Rs 12760
36. An amount of money is to be divided among P, Q and R in the ratio of 3 : 5 : 7, respectively. If the amount received by R is Rs 4,000 more than the amount received by Q, what will be the total amount received by P and Q together?
(a) Rs 8,000 (b) Rs 12,000
(c) Rs 16,000 (d) Cannot be determined
37. A cat takes 5 leaps for every 4 leaps of a dog but 3 leaps of the dog are equal to 4 leaps of the cat. What is the ratio of the speeds of the cat to that of the dog?
(a) 11 : 15 (b) 15 : 11
(c) 16 : 15 (d) 15 : 16
38. Nandita scores 80% marks in five subjects together, viz, Hindi, Science, Mathematics, English and Sanskrit, where in the maximum marks of each subject were 105. How many marks did Nandita score in Science, if she scored 89 marks in Hindi, 92 marks in Sanskrit, 98 marks in Mathematics and 81 marks in English?
(a) 60 (b) 75
(c) 65 (d) 70
39. Out of 120 applications for a post, 70 are males and 80 have a driver's license. What is the ratio between the minimum to maximum number of males having driver's license?
(a) 1 : 2 (b) 2 : 3
(c) 3 : 7 (d) 5 : 7
40. In a factory, the ratio of the numbers of employees of three types A, B and C is 9 : 13 : 18 and their wages are in the ratio of 10 : 7 : 4. If number of employees of type C is 54 and wages of every employee of type B is Rs 1400, then find the total wages of all the employees of type A?
(a) Rs 51000 (b) Rs 54000
(c) Rs 56000 (d) Rs 57000

Foundation

Solutions

1. (d); Let present ages of Ramu and Somu be R and S respectively.

$$\begin{array}{ccc}
 (R-1) & : & (S-1) \Leftrightarrow 6:7 \\
 \uparrow & : & \uparrow \\
 \text{One year ago} \rightarrow & & \\
 R & : & S \\
 \downarrow & : & \downarrow \text{Four years hence} \\
 (R+4) & : & (S+4) \Leftrightarrow 7:8
 \end{array}$$

$$\begin{array}{l}
 \frac{R-1}{S-1} = \frac{6}{7} \quad \left| \quad \frac{R+4}{S+4} = \frac{7}{8} \right. \\
 7R-7 = 6S-6 \quad \left| \quad 8R+32 = 7S+28 \right. \\
 7R-6S = 1 \quad \left| \quad 8R-7S = -4 \right.
 \end{array}$$

$$7R - 6S = 1 \quad \dots (i)$$

$$8R - 7S = -4 \quad \dots (ii)$$

Multiplying equation (i) by 8 and equation (ii) by 7, we get $\rightarrow$

$$56R - 48S = 8 \quad \dots (iii)$$

$$56R - 49S = -28 \quad \dots (iv)$$

Subtract (iv) from (iii) : $S = 36$ years

Shortcut:

One year ago $\rightarrow R : S \Leftrightarrow 6 : 7 \Rightarrow S = 7$

Only option (d) has a number from which, if we subtract '1', we get a number divisible by 7.

2. (b); $33\% \text{ of } A = \frac{33 \times A}{100} = \frac{55 \times B}{100} = 55\% \text{ of } B$

$$= \frac{33 \times A}{100} = \frac{55 \times B}{100} \Rightarrow \frac{A}{B} = \frac{55}{33}$$

$$A : B \Leftrightarrow 5 : 3$$

3. (a); $68\% \text{ of two fifth of } 550$

$$= \frac{68 \times \left(\frac{2}{5} \times 550 \right)}{100} = \frac{68 \times 220}{100} = \frac{748}{5} = 149.6$$

4. (c); Let, number is 'n'

$$\left(\frac{45n}{100} \right) - 24 = 48 \Rightarrow \frac{9n}{20} - 24 = 48$$

$$9n - 480 = 48 \times 20 \Rightarrow 9n = 960 + 480$$

$$n = \frac{1440}{9} = 160 \Rightarrow \frac{3}{8}n = \frac{3}{8} \times 160 = 60$$

5. (a); $30\% \text{ of the number} = 190.8$

$$1\% \text{ of that number} = \frac{190.8}{30}$$

$$175\% \text{ of the number} = \frac{1908}{300} \times 175 = 159 \times 7 = 1113$$

6. (c); $\frac{3}{8} \times 1000 = 375 \Rightarrow 32\% \text{ of } \left(\frac{3}{8} \text{th of } 1000 \right)$

$$= 32\% \text{ of } 375 = \frac{32 \times 375}{100} = 120$$

7. (c); Let the third number be n.

$$20\% \text{ more than } n = n + \frac{n}{5} = \frac{6n}{5}$$

$$50\% \text{ more than } n = n + \frac{n}{2} = \frac{3n}{2}$$

$$\text{Ratio of } \frac{6n}{5} \text{ and } \frac{3n}{2} \Rightarrow \frac{6n}{5} \times \frac{2}{3n} = 4 : 5$$

Shortcut

Let the third number be 100

Then the two numbers are,

20% more and 50% more

$$\begin{array}{cc}
 \downarrow & \downarrow \\
 120 & 150
 \end{array}$$

$$\text{Ratio} = 120 : 150 = 4 : 5$$

8. (c); $A : B : C : D \Rightarrow 5 : 2 : 4 : 3$

$\Rightarrow$ C gets 1 unit more than D.

Given, C gets Rs. 1000 more than D.

$\Rightarrow$ 1 unit = Rs, 1000 $\Rightarrow$ B's share = 2 units
 $= 2 \times 1000 = \text{Rs. } 2000$

9. (b); $0.75 : x :: 5 : 8$

$$\frac{0.75}{x} = \frac{5}{8} \Rightarrow 5x = 6.00 \Rightarrow x = 1.2$$

Shortcut

$$0.75 : x :: 5 : 8 \Rightarrow 0.75 \rightarrow 5 \Rightarrow 1 \rightarrow 0.15 \Rightarrow 8 \rightarrow 1.20$$

10. (b); Three numbers = I, II, III

$$I + II + III = 98, I : II \rightarrow 2 : 3, II : III \rightarrow 5 : 8$$

$$10 : 15 \leftarrow (2 : 3) \times 5 \leftarrow (I : II) \times 5$$

$$15 : 24 \leftarrow (5 : 8) \times 3 \leftarrow (II : III) \times 3$$

$$I : II : III, 10 : 15 : 24, 10 + 15 + 24 = 49$$

$$\text{Given } I + II + III = 98, 98 \div 49 = 2$$

$$\Rightarrow 1 \text{ unit} = 2$$

$$\Rightarrow \text{Second number} = 15 \times 2 = 30$$

11. (c); $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$

L.C.M. of 2, 3 and 4 = 12

On multiplying by a number throughout, the ratio remains same. Here, we multiply by 12.

$$\frac{1}{2} \times 12 : \frac{2}{3} \times 12 : \frac{3}{4} \times 12$$

$$6 : 8 : 9 \Rightarrow 6 + 8 + 9 = 23$$

$$23 \text{ units} = 872 \Rightarrow 1 \text{ unit} = 37.91$$

$$\Rightarrow 1\text{st part} = 6 \text{ units} = 6 \times 37.91 = 227.46$$

12. (b); Fourth proportional to 5, 8, 15

$$5 : 8 :: 15 : ?$$

$$\frac{5}{8} = \frac{15}{x} \Rightarrow x = 24$$

Shortcut

$$5 \xrightarrow{\times 3} 15$$

$$8 \xrightarrow{\times 3} 24$$

13. (b); Let the two numbers be a and b

$$\frac{a}{b} = \frac{3}{5} \Rightarrow 5a = 3b$$

... (i)

$$\frac{a-9}{b-9} = \frac{12}{23} \Rightarrow 23a - 207 = 12b - 108$$

$$\Rightarrow 23a - 12b = 99$$

... (ii)

$$b = \frac{5a}{3} \text{ From (i)}$$

... (iii)

Using (ii) and (iii)

$$23a - 12 \times \frac{5a}{3} = 99 \Rightarrow 3a = 99 \Rightarrow a = 33$$

Shortcut

Given = Two numbers are in the ratio

$$= 3 : 5$$

↓

Smaller number must be a multiple of 3

Only options (a) and (b) have '3' as a factor.

Given : On subtracting 9 from each, the two numbers are in the ratio 12 : 23

On subtracting 9 from 27 → option (a),

We get 18, but the ratio becomes 12 : 23,

So, option (a) → discarded.

On subtracting 9 from 33 → option (b),

We get 24

↓

multiple of 12.

14. (d); Let the two numbers be a and b.

$$a - (40\% \text{ of } a) = \frac{2}{3}b$$

$$a - \frac{40a}{100} = \frac{2}{3}b \Rightarrow \frac{60a}{100} = \frac{2}{3}b$$

$$\frac{b}{a} = \frac{60 \times 3}{2 \times 100} = \frac{9}{10}$$

⇒ Ratio of the second number to the first number = 9 : 10

$$15. (c); \frac{5a+3b}{2a-3b} = \frac{23}{5}$$

Dividing numerator and denominator by 'b'

$$\frac{5\left(\frac{a}{b}\right)+3}{2\left(\frac{a}{b}\right)-3} = \frac{23}{5}$$

$$\text{Let } \frac{a}{b} = x \Rightarrow \frac{5x+3}{2x-3} = \frac{23}{5}$$

$$25x + 15 = 46x - 69 \Rightarrow 21x = 84$$

$$x = 4 \Rightarrow \frac{a}{b} = 4 \Rightarrow a : b \Leftrightarrow 4 : 1$$

16. (c); P : Q = 8 : 15

$$P : Q : R = ?$$

$$Q : R = 3 : 2) \times 5 \Rightarrow Q : R = 15 : 10$$

$$P : Q = 8 : 15 \Rightarrow 15 : 10 = Q : R$$

$$P : Q : R = 8 : 15 : 10$$

17. (a); P : Q = 8 : 15

... (i)

$$Q : R = 5 : 8) \times 3 \Rightarrow Q : R = 15 : 24$$

... (ii)

From (1) and (2)

$$P : Q : R = 8 : 15 : 24$$

... (iii)

$$R : S = 4 : 5) \times 6 \Rightarrow R : S \Leftrightarrow 24 : 30$$

... (iv)

$$P : Q : R = 8 : 15 : 24 \Rightarrow 24 : 30 = R : S$$

$$P : Q : R : S = 8 : 15 : 24 : 30 \Rightarrow P : S = 8 : 30 = 4 : 15$$

18. (a); Let the fourth proportional to 4, 16, 7 be x.

$$4 : 16 :: 7 : x$$

$$\frac{4}{16} = \frac{7}{x} \Rightarrow 4x = 16 \times 7 \Rightarrow x = 4 \times 7 = 28$$

19. (b); Let the mean proportion be = r.

$$9 : r :: r : 64 \Rightarrow \frac{9}{r} = \frac{r}{64}$$

$$r^2 = 9 \times 64 \Rightarrow r = 3 \times 8 = 24$$

20. (a); Duplicate ratio = ratio of squares

Duplicate ratio of

$$2 : 7 = 2^2 : 7^2 = 4 : 49$$

21. (c); Sub-duplicate ratio = ratio of square roots

Sub-duplicate ratio of

$$81 : 64 = \sqrt{81} : \sqrt{64} = 9 : 8$$

22. (b); Triplicate ratio = Ratio of cubes.

Triplicate ratio of

$$7 : 5 = 7^3 : 5^3 = 343 : 125$$

23. (a); Inverse ratio of 17 : 19 = 19 : 17

24. (b); Compound ratio of = 2 : 7, 5 : 3 and 4 : 7 will be

$$2 \times 5 \times 4 : 7 \times 3 \times 7 = 40 : 147$$

25. (d); B : C = 8 : 9

$$A : B = 3 : 4) \times 2 \Rightarrow A : B = 6 : 8$$

$$8 : 9 = B : C \Rightarrow A : B : C = 6 : 8 : 9$$

26. (d); $a : b = 3 : 5 \Rightarrow a : b = 12 : 20$
 $b : c = 4 : 7 \Rightarrow b : c = 20 : 35$
 $\Rightarrow a : b : c = 12 : 20 : 35 \Rightarrow a : c = 12 : 35$

27. (a); $P : Q : R = 2 : 3 : 4$
 $P \rightarrow 2, Q \rightarrow 3, R \rightarrow 4$

$$\frac{P}{Q} = \frac{2}{3}, \frac{Q}{R} = \frac{3}{4}, \frac{R}{P} = \frac{4}{2}$$

$$\frac{P}{Q} : \frac{Q}{R} : \frac{R}{P} = \frac{2}{3} : \frac{3}{4} : \frac{4}{2} = \left(\frac{2}{3} : \frac{3}{4} : \frac{4}{2} \right) \times 12 = 8 : 9 : 24$$

28. (a); $\frac{a}{3} = \frac{b}{8} \Rightarrow \frac{a}{b} = \frac{3}{8}$
 $a \rightarrow 3, b \rightarrow 8, a + 3 \rightarrow 6$

$$b + 8 \rightarrow 16 \Rightarrow \frac{a+3}{b+8} = \frac{6}{16} = \frac{3}{8}$$

29. (a); $4^3 : 2^5 = ?$

$$= (2^2)^3 : 2^5$$

$$= 2^6 : 2^5 = \frac{2^6}{2^5} \Rightarrow 2 : 1$$

30. (b); $2 : x :: 5 : 7$

$$\frac{2}{x} = \frac{5}{7}, x = \frac{2 \times 7}{5} \Rightarrow x = 2.80$$

31. (c); Ratio of salaries $\rightarrow A : B : C = 2 : 3 : 5$
 $A \rightarrow 2x, B \rightarrow 3x, C \rightarrow 5x$

$$A + 15\% \text{ of } A = 2x + \frac{15 \times 2x}{100} = \frac{46x}{20}$$

$$B + 10\% \text{ of } B = 3x + \frac{3x \times 10}{100} = \frac{33x}{10}$$

$$C + 20\% \text{ of } C = 5x + \frac{5x \times 20}{100} = 6x$$

New ratio of their salaries

$$A : B : C = \frac{46x}{20} : \frac{33x}{10} : 6x$$

$\downarrow$ Multiply by 20

$$46x : 66x : 120x \Rightarrow 23x : 33x : 60x \Rightarrow 23 : 33 : 60$$

Shortcut

Ratio of salaries of A, B and C $\rightarrow 2 : 3 : 5$

Let the salaries be

$$200 \quad 300 \quad 500$$

$$\downarrow 15\% \uparrow \quad \downarrow 10\% \uparrow \quad \downarrow 20\% \uparrow$$

$$230 \quad 330 \quad 600$$

$$\text{New ratio} = 230 : 330 : 600 = 23 : 33 : 60$$

32. (d);

$$\begin{array}{c} 782 \\ \swarrow \quad \downarrow \quad \searrow \\ \frac{1}{2} : \frac{2}{3} : \frac{3}{4} \end{array}$$

L.C. M. of 2, 3 and 4 = 12

$$\left(\frac{1}{2} : \frac{2}{3} : \frac{3}{4} \right) \times 12 \Rightarrow 6 : 8 : 9 \Rightarrow 6 + 8 + 9 = 23$$

$$23 \text{ units} = 782 \Rightarrow 1 \text{ unit} = \frac{782}{23} = 34$$

First part = 6 units = $6 \times 34 = 204$

33. (c); Let the two numbers be a and b

$$a : b = 1 : 2$$

$$\frac{a}{b} = \frac{1}{2} \Rightarrow 2a = b \quad \dots (i)$$

$$\frac{a+7}{b+7} = \frac{3}{5} \Rightarrow 5a + 35 = 3b + 21 \quad \dots (ii)$$

From (i) and (ii)

$$5a + 35 = 3 \times 2a + 21 \Rightarrow 5a + 35 = 6a + 21$$

$$a = 14$$

$$\text{Greatest number} = b = 2a = 28$$

Shortcut:

On adding 7 to both, ratio changes to 3 : 5

$\Rightarrow$ On adding 7 to the greatest number, the

resulting number must be a multiple of 5

only option (c) has the number 28, in which, if we add 7, we get 35, which is a multiple of 5.

$$3 : 5 :: ? : 35$$

$$\downarrow$$

$$21$$

$$\Rightarrow \text{Smaller number} = 21 - 7 = 14$$

$$14 : 28 = 1 : 2$$

34. (c); Ratio of 3 numbers is 3 : 4 : 5

Let the numbers be 3x, 4x, 5x

Sum of their squares = 1250

$$9x^2 + 16x^2 + 25x^2 = 1250 \Rightarrow 50x^2 = 1250$$

$$x^2 = 25 \Rightarrow x = 5 \Rightarrow 3x + 4x + 5x = 12x = 60$$

35. (c); Let age of Sachin = S

Age of Rahul = R

$$S = R - 4 \quad \dots (i)$$

$$\frac{S}{R} = \frac{7}{9} \Rightarrow 9S = 7R \quad \dots (ii)$$

From (1)

$$9S = 9R - 36 \Rightarrow 9S = 7R \Rightarrow 7R = 9R - 36$$

$$2R = 36 \Rightarrow R = 18$$

$$\text{Age of Sachin} = S = R - 4 = 18 - 4 = 14 \text{ years}$$

Shortcut:

Sachin : Rahul

S : R

7 : 9

Difference = 2 units = 4 years

$\Rightarrow 1 \text{ unit} = 2 \text{ years}$

Sachin's age = 7 units = $7 \times 2 = 14 \text{ years}$

36. (b); Arun : Deepak

A : D = 4 : 3

Let A = $4x \Rightarrow D = 3x$

After 6 years, Arun's age = $4x + 6$

Deepak's age = $3x + 6$

$4x + 6 = 26$ (given)

$4x = 20 \Rightarrow x = 5$

Deepak's present age = $3x = 3 \times 5 = 15 \text{ years}$.

Shortcut

After 6 years, Arun's age will be 26 years.

$\Rightarrow$ Present age of Arun = 20 years

Ratio of present ages = Arun : Deepak

4	:	3
$\downarrow \times 5$		$\downarrow \times 5$
20 years		15 years

37. (a); Present Ages $\Rightarrow X : Y$

5 : 6

Let the ages be $5x$ and $6x$

$\downarrow$	$\downarrow$	7 yrs. hence
$5x + 7$		$6x + 7$

$$\frac{5x+7}{6x+7} = \frac{6}{7}$$

$$\Rightarrow 35x + 49 = 36x + 42 \Rightarrow x = 7$$

X's present age = $5x = 5 \times 7 = 35 \text{ years}$.

Shortcut

Present ages 5 : 6

$\downarrow + 1$ $\downarrow + 1 = 1 \text{ unit} = 7$

After 7 years 6 : 7

$\Rightarrow$ X's present age = $7 \times 5 = 35 \text{ years}$

38. (a); Ratio of present ages $\rightarrow$ Sameer : Anand

5 : 4

Let present ages be $5x$ and $4x$

Three years hence $\rightarrow (5x + 3)$ and $(4x + 3)$

$$\frac{5x+3}{4x+3} = \frac{11}{9} \Rightarrow 45x + 27 = 44x + 33 \Rightarrow x = 6$$

Anand's present age = $4x = 4 \times 6 = 24$

Shortcut

Ratio of present age S : A

5 : 4

Anand's present age must be a multiple of 4.

Check option (a) = 24

Sameer : Anand

$\downarrow$	:	$\downarrow$
5	:	4
$\downarrow \times 6$		$\downarrow \times 6$
30		24
$\downarrow + 3$		$\downarrow + 3$
33	:	27
$\updownarrow$		

11 : 9 Satisfied

39. (d); Ten years ago, Ratio of ages of Jayant, Prem and Saransh $\rightarrow J : P : S$

2 : 3 : 4

Let the ages be $2x$ $3x$ $4x$

$\downarrow$ $\downarrow$ $\downarrow$

Present ages $\rightarrow (2x + 10) (3x + 10) (4x + 10)$

Total of the present ages = 93 years.

$$\Rightarrow 2x + 10 + 3x + 10 + 4x + 10 = 93$$

$$9x + 30 = 93 \Rightarrow x = 7$$

$$\Rightarrow \text{Present age of Saransh} = 4x + 10$$

$$= 4 \times 7 + 10 = 38 \text{ years.}$$

Shortcut:

On subtracting 10 from the present age of Saransh, we must get a multiple of 4, because ten years ago, the ratio of ages of Jayant, Prem and Saransh was 2 : 3 : 4.

From the option $\rightarrow$ (a) and (d) can be discarded.

(a) 24 yrs $\xrightarrow{-10}$ 14 yrs.
(not a multiple of 4.)

(b) 32 yrs. $\xrightarrow{-10}$ 22 yrs.
(not a multiple of 4)

Check option (c) 34 yrs.

10 years ago $34 - 10 = 24 \text{ yrs.}$

J	:	P	:	S
2	:	3	:	4
$\downarrow \times 6$		$\downarrow \times 6$		$\downarrow \times 6$
12		18		24
$\downarrow + 10$		$\downarrow + 10$		$\downarrow + 10$
22	+	28	+	34 $\neq$ 93

Check (d) 38 yrs.

$38 - 10 = 28 \text{ yrs.}$

2	:	3	:	4
$\downarrow \times 7$		$\downarrow \times 7$		$\downarrow \times 7$
14		21		28
$\downarrow + 10$		$\downarrow + 10$		$\downarrow + 10$
24	+	31	+	38 = 93 yrs.

40. (b); Hitesh Ronnie

↓ ↓
40 yrs. 60 yrs.

It is clear from option (b) that 10 yrs. ago their ages were 30 yrs. and 50 yrs., i.e., in the ratio 3 : 5

41. (b); Ratio of profits = ratio of investments
= 45000 : 30000 = 45 : 30 = 3 : 2

42. (a); Since the investments were made for the same period of time, ratio of shares of profit = ratio of investments = 60000 : 50000 = 6 : 5

Kishor's share = $\frac{5}{11} \times \text{total profit} = \frac{5}{11} \times 33000$
= 5 × 3000 = 15000

43. (d); Investments

↓
Anil Rs. → 3300

Nikhil Rs. → 5000

Ratio of profit shares = ratio of investments
= 3300 : 5000

Ratio of profit shares = 33 : 50

↓ ↓
Anil Nikhil

Profit earned = 1660 Rs.

Anil's share = $\frac{33}{83} \times 1660 = \text{Rs. } 660$

44. (b); Rahul invested for one year i.e., 12 months. 6 months later, Sanjay joined him. So, Sanjay invested for 6 months only.

	Rahul	Sanjay
(I) Investment →	45000	30000
(T) Time period →	12 months	6 months

↓ ↓

Product → 45000 × 12 30000 × 6

Ratio of profits = 45000 × 12 : 30000 × 6 = 3 : 1

45. (b); Ratio of investments = Ratio of profit shares
= 6 : 5 : 8

↓ ↓ ↓
Anil : Nikhil : Sourabh
Profit earned = Rs. 83,600

Saurabh's share = $\frac{8}{19} \times 83600 = 35,200 \text{ Rs.}$

46. (c);

	Ram	Karan	Rohan
Ratio of capitals	2	3	4
Ratio of time	6	4	3
Ratio of profits	Ram	Karan	Rohan
	↓	↓	↓
	2 × 6	3 × 4	4 × 3
	12	12	12
	1	1	1

47. (d);

	Ravi	Kavi	
Investments	8000	72000	(Rs.)

(Ratio of)
Investments 8 : 72

1 : 9

Ratio of investments = Ratio of profits = 1 : 9

48. (b);

	Srikant	Vividh
Investments	185000	225000
Ratio	185 : 225	
	37 : 45	

Let the total profit be P

Vividh's share = $\frac{45}{45+37} \times P = 9000$

$\frac{45}{82} \times P = 9000$

$P = \frac{82 \times 9000}{45} = 82 \times 200 = 16400$

49. (d);

	Rajan	Sajan
Investments →	Rs. 14200	Rs. 15600
Ratio →	142 : 156	
	⇕	
	71 : 78	

Total profit = Rs. 74500

Rajan's share = $\frac{71}{149} \times 74500 = 35500$

50. (a);

	A	B
Investments →	Rs. 4000	Rs. 16000
		⇕
Ratio of investments	1	4
Ratio of times periods	12 months	9 months
		⇕
	4	3
Ratio of profits :	(1 × 4)	(4 × 3)
	= 4	12
		⇕
	1	3

Moderate

1. (b); Let the ten's digit and the unit's digit of the number be x and y respectively.

The number is $\rightarrow 10x + y$

on interchanging digits $\rightarrow 10y + x$

$$x : y = 1 : 2 \rightarrow y = 2x$$

$$\begin{aligned} \Rightarrow 10x + y &= 12x \\ 10y + x &= 21x \end{aligned} \rightarrow [\text{difference} = 36 (\text{given})]$$

$$21x - 12x = 36$$

$$9x = 36 \Rightarrow x = 4$$

$\Rightarrow$ The number is $12x$

$$\Rightarrow 12x = 12 \times 4 = 48$$

on interchanging $\rightarrow 21x = 21 \times 4 = 84$

$$\text{Sum of digits} = x + y = 12$$

$$\text{difference of digits} = y - x = 4$$

$$\text{difference of sum and difference of digits} = 12 - 4 = 8$$

2. (a);
- | | | |
|----------------------|----------------------|----------------------|
| Maths | Phy. | Bio. |
| $\downarrow$ | $\downarrow$ | $\downarrow$ |
| 500 | 700 | 800 |
| 40% inc $\downarrow$ | $\downarrow$ 50% inc | $\downarrow$ 75% inc |
| 700 | 1050 | 1400 |
- New ratio = $70 : 105 : 140 = 2 : 3 : 4$

3. (b);
- 60 litres

milk

$\downarrow$

40 L

water

$\downarrow$

20 L

$\rightarrow 3 \text{ units} = 60 \text{ L}$
 $\Rightarrow 1 \text{ unit} = 20 \text{ L}$

1 : 3] $\times 2$ (Multiply by 2, to keep milk constant)

$\downarrow$

2 : 6

$\downarrow$

6 units = $6 \times 20 = 120 \text{ L}$

So, $120 - 20 = 100 \text{ L}$

4. (c);
- | | |
|----------------------|----------------------|
| Boys | Girls |
| $\downarrow$ | $\downarrow$ |
| 7 | 8 |
| 700 | 800 |
| 20% inc $\downarrow$ | $\downarrow$ 10% inc |
| 840 | 880 |
- New ratio = $840 : 880 = 21 : 22$

5. (d);

Ravi	:	Sumit
2	:	3
Let $\rightarrow 2x$	,	$3x$
$+4000 \downarrow$		$\downarrow +4000$
$(2x+4000)$		$(3x+4000)$
$\frac{2x+4000}{3x+4000} = \frac{40}{57}$		

$$114x + 228000 = 120x + 160000$$

$$6x = 68000 \Rightarrow x = \frac{68000}{6}$$

$$\text{Sumit's salary} = 3x = 3 \times \frac{68000}{6} = \text{Rs. } 34000$$

6. (c);

	A	B	C
Ratio of salaries	2	3	5
Let the salaries be	200	300	500
	$\downarrow 15\%$	$\downarrow 10\%$	$\downarrow 20\%$
	230	330	600

New ratio $\rightarrow 23 : 33 : 60$

7. (d); 25p, 10p and 5p coins are in the ratio of $2 : 3 : 4$.

So, number of 25p coins = 2 units

Its value = $25 \times 2 = 50$ units

number of 10 p coins = 3 units

Its value = $3 \times 10 = 30$ units

number of 5p coins = 4 units

Its value = $4 \times 5 = 20$ units

Total value = $50 + 30 + 20 = 100$ units = 50 Rs.

$$1 \text{ unit} = \frac{1}{2} \text{ Rs.} \Rightarrow 1 \text{ unit} = 50 \text{ p}$$

value of 5p coins = 20 units = $20 \times 50 \text{ p}$.

$$\Rightarrow \text{Number of 5p coins} = \frac{20 \times 50}{5} = 200$$

8. (d);

sum 1	sum 2
C : A : B 4 : 5 : 6 $4x : 5x : 6x$ $6x - y = 2000$ M : N 1 : 1 y : y $4x = ?$	

Can't determine x from this equation

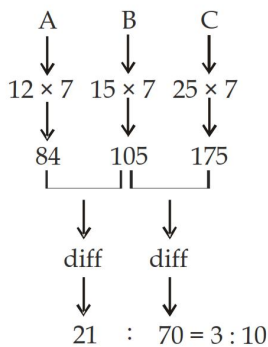
9. (b);

$$A : B : C \Rightarrow 12 : 15 : 25$$

Let $\rightarrow 12x, 15x, 25x$

$$\text{Sum} = 52x = 364$$

$$x = \frac{364}{52} \Rightarrow x = 7$$



10. (b); $\frac{625}{x} = \frac{x}{1156}$
 $x^2 = 625 \times 1156 \Rightarrow x^2 = (25)^2 \times (34)^2$
 $x = 25 \times 34 = 850$

11. (c);

Initial ratio $\rightarrow$ Alcohol: Water
 New ratio $\rightarrow$ 4 : 3 $\rightarrow$ 2 unit increase = 5 L
 4 : 5 $\leftarrow$ 1 unit = 2.5 L
 Alcohol in initial mixture = 4 units
 $= 4 \times 2.5 = 10$ L

12. (c); $\frac{25p}{1} : \frac{10p}{2} : \frac{5p}{3}$
 $\downarrow \quad \downarrow \quad \downarrow$
 $1 \times 25 \quad 2 \times 10 \quad 3 \times 5$
 $25p \quad 20p \quad 15p$
 $\swarrow \quad \downarrow \quad \searrow$
 Total = 60 units
 $= 30 \text{ Rs} = 3000 \text{ p}$

1 unit = 50p
 value of 5p coins = 15 units = 750p

Number of 5p coins = $\frac{750}{5} = 150$

13. (b); Let C's share be $\rightarrow C$

Then, B's share = $\frac{1}{4} C$

A's share = $\frac{2}{3} \times \frac{1}{4} C = \frac{1}{6} C$

Total = Rs. 510 $\Rightarrow \frac{C}{6} + \frac{C}{4} + C = 510$

$\frac{2C + 3C + 12C}{12} = 510 \Rightarrow 17C = 12 \times 510$

$C = 12 \times 30 = 360$

So, C's share = Rs. 360

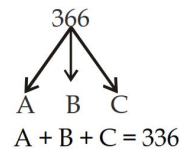
B's share = Rs. 90

A's share = Rs. 60

Shortcut: only option (b) satisfies the given conditions.

14. (a); Let $\rightarrow$ A's share be A

B's share be B and C's share be C



$A + B + C = 336$

... (i)

$A = \frac{1}{2}(B + C)$

$2A = B + C$

... (ii)

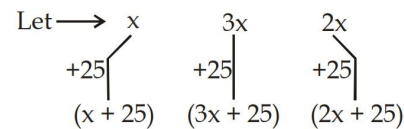
From (1) and (2)

$A + 2A = 366 \Rightarrow A = 122$

A's share = Rs. 122

15. (d); Data inadequate

16. (d); Ratio $\rightarrow 1 : 3 : 2$



$x + 25 + 3x + 25 + 2x + 25 = 735$

$6x + 75 = 735 \Rightarrow 6x = 660 \Rightarrow x = 110$

Money received by C = $2x + 25$

$= 2 \times 110 + 25 = \text{Rs. } 245$

17. (d); Ratio of remainders A B C
 3 : 4 : 5

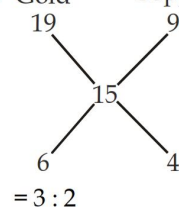
Let $\rightarrow$ 3x 4x 5x

Sum of original shares $\rightarrow (3x + 5) + (4x + 10) + (5x + 15) = 2430 \Rightarrow 12x + 30 = 2430$

$12x = 2400 \Rightarrow x = 200$

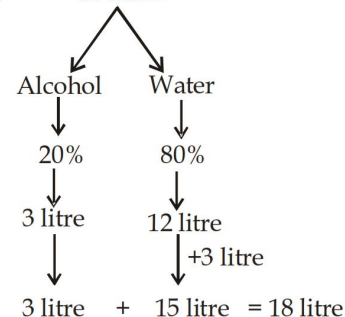
B's share = $4x + 10 = 810 \text{ Rs.}$

18. (d); Gold Copper



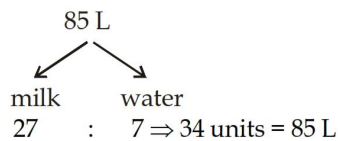
$= 3 : 2$

19. (b); 15 litres



alcohol = $\frac{3}{18} \times 100 = \frac{50}{3} \% = 16\frac{2}{3} \%$

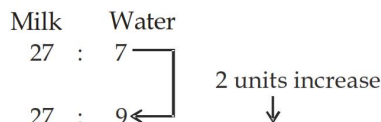
20. (a);



$$\Rightarrow 1 \text{ unit} = \frac{85}{34} \text{ L} = \frac{5}{2} \text{ L}$$

New ratio $\rightarrow [3 : 1] \times 9$ [because. water is added]

New ratio $\rightarrow 27 : 9$



$$= 2 \times \frac{5}{2} = 5 \text{ L water}$$

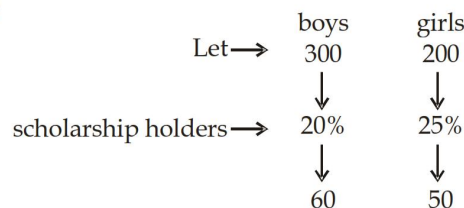
21. (b); Ratio of sides $\rightarrow \left(\frac{1}{2} : \frac{1}{3} : \frac{1}{4}\right) \times 12 = 6 : 4 : 3$

perimeter = $6 + 4 + 3 = 13$ units = 104 cm

1 unit = 8 cm

Longest side $\rightarrow 6$ units = $6 \times 8 = 48$ cm

22. (c);



Students who got scholarship $\rightarrow 60 + 50 = 110$

students who did not get scholarship

$\rightarrow 500 - 110 = 390$

$$\text{Percentage of student} = \frac{390}{500} \times 100 = 78\%$$

23. (c); Change the ratios into fractions :

Containers	Milk	Water	Volume of containers
I	$\frac{4}{5}$	$\frac{1}{5}$	$\frac{3}{12} = \frac{1}{4}$
II	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{4}{12} = \frac{1}{3}$
III	$\frac{5}{7}$	$\frac{2}{7}$	$\frac{5}{12}$

Ratio of milk and water in the fourth container is:

$$\frac{4}{5} \times \frac{1}{4} + \frac{3}{4} \times \frac{1}{3} + \frac{5}{7} \times \frac{2}{7} : \frac{1}{5} \times \frac{1}{4} + \frac{1}{4} \times \frac{1}{3} + \frac{2}{7} \times \frac{5}{12}$$

$$= \frac{1}{5} + \frac{1}{4} + \frac{25}{84} : \frac{1}{20} + \frac{1}{12} + \frac{5}{42} = \frac{314}{420} : \frac{106}{420}$$

$$= 314 : 106 = 157 : 53$$

24. (d); Let, $x = \frac{K}{y^2}$ [K = constant]

$$1 = \frac{K}{4} \Rightarrow K = 4 \Rightarrow x = \frac{4}{y^2}$$

$$y = 6 \Rightarrow x = \frac{4}{36} = \frac{1}{9}$$

25. (a); Let the fixed amount be Rs. x and the cost of each unit be Rs. y

$$\text{Then, } 500y + x = 1680 \quad \dots (i)$$

$$600y + x = 1980 \quad \dots (ii)$$

Subtract (i) from (ii)

$$100y = 300 \Rightarrow y = 3$$

Putting $y = 3$ in (i)

$$1500 + x = 1680 \Rightarrow x = 180$$

So, fixed charges = Rs. 180

Charge per unit = Rs. 3.

Total chargers for Consuming 700 units

$$= 700 \times 3 + 180 = 2100 + 180 = 2280$$

26. (c);

	A	B
Income $\rightarrow$	5x	4x
Expenditure $\rightarrow$	3y	2y
Saving $\rightarrow$	(5x - 3y)	(4x - 2y)
	$5x - 3y = 1600 \times 2$	$4x - 2y = 1600 \times 3$
	$10x - 6y = 3200$ (1)	$12x - 6y = 4800$ (2)
	From (1) and (2) $2x = 1600 \Rightarrow x = 800$	
	A's income = $5x = 5 \times 800 = 4000$ Rs.	

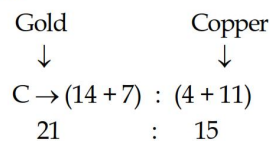
27. (c); Gold : copper

$$A \rightarrow 7 : 2 \times 2$$

$$A \rightarrow 14 : 4 \rightarrow \text{total} = 14 + 4 = 18$$

$$B \rightarrow 7 : 11 \rightarrow \text{total} = 7 + 11 = 18$$

18 units of A and 18 units of B are melted to form C



Ratio $\rightarrow 7 : 5$

28. (a); Let present ages be $\rightarrow P \quad Q$

$$\rightarrow 5x \quad 7x$$

$$P's \text{ age after 6 yrs.} = 5x + 6$$

$$7x - (5x + 6) = 2 \Rightarrow 2x - 6 = 2$$

$$2x = 8 \Rightarrow x = 4$$

So, present ages of P and Q are 20 years & 28 years respectively.

Total = 48 years

- $$7x \times 3x = 756 \Rightarrow 21x^2 = 756$$
- $$x^2 = 36 \Rightarrow x = 6$$
- $$\Rightarrow \text{Present ages} \rightarrow \text{father} \rightarrow 7 \times 6 = 42 \text{ yrs.}$$
- $$\text{son} \rightarrow 3 \times 6 = 18 \text{ yrs.}$$
- $$\text{After 6 years} \rightarrow \text{father} \rightarrow 48 \text{ years.}$$
- $$\text{son} \rightarrow 24 \text{ yrs.}$$

- So, sum of their present ages = $56 + 24 = 80$ years
Proportion $\rightarrow 4 : 7 : 9$

So, the present ages are $\rightarrow \frac{4}{20} \times 80 = 16$ Years
 $\frac{7}{20} \times 80 = 28$ Years
 $\frac{9}{20} \times 80 = 36$ Years

- Let the ages be → man → $4x$
 wife → $3x$
- After 4 yrs → man → $4x + 4$
 wife → $3x + 4$
- $$\frac{4x+4}{3x+4} = \frac{9}{7}$$
- $$28x + 28 = 27x + 36 \Rightarrow x = 8$$
- Present ages → man → 32 yrs.
 wife → 24 yrs.
- From the option if we check option (c) → 12 yrs.
 then 12 yrs. ago → man → 20 yrs.
 wife → 12 yrs.

- | | A | B |
|-------------------------------|----------------------------|----------|
| Present ages $\rightarrow$ | $5x$ | $3x$ |
| A's age 4 yrs. ago = | $5x - 4$ | |
| B's age 4 yrs. hence = | | $3x + 4$ |
| $5x - 4 = 3x + 4 \Rightarrow$ | $2x = 8 \Rightarrow$ | $x = 4$ |
| A's age 4 yrs. hence = | $5 \times 4 + 4 = 24$ yrs. | |
| B's age 4 yrs. ago = | $3 \times 4 - 4 = 8$ yrs. | |
| Ratio $\rightarrow$ | $24 : 8 = 3 : 1$ | |

- present age of the person = $\frac{2}{5}M$
 After 8 yrs. $\rightarrow$ mother's age = $M + 8$
 person's age = $\frac{2}{5}M + 8$
 $\frac{2}{5}M + 8 = \frac{M}{2} + 4, \quad \frac{M}{10} = 4 \Rightarrow M = 40$

- | | Father | Son |
|-------------------------|--------------------------|-----------|
| Present ages | f | s |
| 4 years ago | $(f - 4)$ | $(s - 4)$ |
| After 4 years | $(f + 4)$ | $(s + 4)$ |
| $f - 4 = 3(s - 4)$ | $(f + 4) + (s + 4) = 64$ | |
| $f - 4 = 3s - 12$ | $f + s = 64 - 8$ | |
| $f - 3s = -8 \dots (i)$ | $f + s = 56 \dots (ii)$ | |

- $$3f + 3s = 168$$
- $$f - 3s = -8$$
- On solving (i) and (ii)
- $$4f = 160 \Rightarrow f = 40$$
- So, father's age at present = 40 years.

- | | | |
|----------------|---------|-------------------|
| | Usha | Reena |
| Present ages → | 36 yrs. | 24 yrs. |
| | ↓ -8 | ↓ -8 (8 yrs. ago) |
| | 28 yrs. | 16 yrs. |
| Ratio → | 28 : 16 | (Usha : Reena) |
| = 7 : 4 | | |

- Ratio of present ages
- ↓
- | | | |
|------|---|--------|
| Anil | : | Poorvi |
| 3 | : | 2 |
- Let →
- | | | |
|----|--|----|
| 3x | | 2x |
| ↓ | | ↓ |
- After 8 yrs. → $(3x + 8)$ $(2x + 8)$
- $$\frac{3x+8}{2x+8} = \frac{25}{18} \Rightarrow 54x + 144 = 50x + 200$$
- $$4x = 56 \Rightarrow x = 14$$
- ⇒ Poorvi's present age = $2x = 28$ yrs.
- Shortcut :**
- Check → option (b)
- Poorvi's present age = 28 yrs.
- $$\Rightarrow \text{Anil's present age} = \frac{3}{2} \times 28 = 42 \text{ yrs.}$$

- $$\Rightarrow \text{Anil's present age} = \frac{3}{2} \times 28 = 42 \text{ yrs.}$$

- After 8 yrs. $\rightarrow$
 Anil $\rightarrow$ 50 yrs.
 Poorvi $\rightarrow$ 36 yrs.
 ratio $\rightarrow$ Anil : Poorvi
 50 : 36
 25 : 18

- Hence verified

- Ratio of ages $\downarrow$
- | | | |
|-------|---|-------|
| Meena | : | Seema |
| 7 | : | 8 |
- $\downarrow$
- difference = 1 unit = 3 yrs.
- Sum = $7 + 8 = 15$ units = $15 \times 3 = 45$ years.

38. (a); Rehana's age after 7 yrs. = 85 yrs.
 $\Rightarrow$ Rehana's present age = $85 - 7 = 78$ yrs.
 Wasim is $\rightarrow$ 12 younger than Rehana
 $\Rightarrow$ Wasim's present age = 66 yrs

Ratio of present ages $\rightarrow$

Manoj	:	Wasim
3	:	11
$\downarrow \times 6$		$\downarrow \times 6$
18 yrs.		66 yrs.

$\downarrow$
 Present age of Manoj's father = $18 + 25 = 43$ yrs.

39. (b); Ratio of present ages $\rightarrow$

Indira	:	Lizzy
3	:	8
$\downarrow$		$\downarrow$
$3x$		$8x$

Let $\rightarrow$

After 8 yrs. $\rightarrow$

$3x + 8$	$8x + 8$
----------	----------

$3x + 8 = 20 \Rightarrow 3x = 12 \Rightarrow x = 4$
 $\Rightarrow$ Lizzy's present age $\Rightarrow 8x = 8 \times 4 = 32$ yrs.
 $\Rightarrow$ Lizzy's age 5 yrs. ago $\Rightarrow 32 - 5 = 27$ yrs.

Shortcut:

Indira	:	Lizzy
3	:	8

Indira's age after 8 yrs. = 20 yrs.
 $\Rightarrow$ Indira's present age = 12 yrs.
 3 units $\rightarrow$ 12 years
 $\Rightarrow$ 1 unit = 4 years
 Lizzy's present age = 8 units = $8 \times 4 = 32$ yrs.
 Lizzy's age 5 yrs. ago = $32 - 5 = 27$ yrs.

40. (d); Let Sarita's present age = s

Kavita's present age = 2s
 8 years. hence
 Sarita's age = s + 8
 Kavita's age = 2s + 8

$$\frac{2s + 8}{s + 8} = \frac{22}{13}$$

$$26s + 104 = 22s + 176$$

$$4s = 72$$

$$s = 18 \Rightarrow \text{Kavita's present age} = 2s = 36 \text{ years}$$

Shortcut

Check option (d)

Kavita's present age = 36 yr.

Sarita's present age = 18 yrs.

Sarita	Kavita
18	36
$\downarrow + 8$	$\downarrow + 8$
26	44

Ratio = 13 : 22

Hence verified

41. (c); Age of A and B 10 years ago
- | | |
|-------|-------|
| A | B |
| $13x$ | $17x$ |

Ratio of ages of A and B after 17 years

$$\frac{13x + 27}{17x + 27} = \frac{10}{11}$$

$$143x + 297 = 170x + 270$$

$$27x = 27, x = 1$$

$$\text{Present age of B} = 17x + 10$$

$$= 17 \times 1 + 10 = 27 \text{ years}$$

42. (d); Can't be determined.

43. (c);

Ram	Sanjay	Aditya
$\downarrow$	$\downarrow$	$\downarrow$
45000	60000	90,000
12 months	9 months	6 months
Ram	Sanjay	Aditya
$\downarrow$	$\downarrow$	$\downarrow$

Ratio of Profits $\rightarrow$

$$45000 \times 12 : 60000 \times 9 : 90000 \times 6$$

$$5400 : 5400 : 5400$$

$$1 : 1 : 1$$

Net profit = Rs. 16500

$$\text{Ram's share} = \frac{1}{3} \times 16500 = \text{Rs. } 5500$$

44. (c);

Ram	Anil	Aditya
$\downarrow$	$\downarrow$	$\downarrow$
1200	1500	900
12 months	6 months	9 months
	Ram	Anil
	$\downarrow$	$\downarrow$

Ratio of profits

$$14400 : 9000 : 8100$$

$$144 : 90 : 81$$

$$16 : 10 : 9$$

Total profit = Rs. 2450

$$\text{Aditya's share} = \frac{9}{35} \times 2450 = \text{Rs. } 630$$

45. (d); Let number of 10-paise coins be 17 k
 and number of 25-paise coins be 6 k.

$$17k \times 0.10 + 6k \times 0.25 = 112$$

$$\Rightarrow 1.7k + 1.5k = 112$$

$$\Rightarrow 3.2k = 112$$

$$\Rightarrow k = 35$$

$$\therefore \text{Number of 10-paise coins} = 17 \times 35 = 595$$

46. (c);
- | | | |
|----------------|------------------|----------------|
| P | Q | S |
| ↓ | ↓ | ↓ |
| (50, 4 months) | (45, 5 months) | (70, 7 months) |
| (25, 8 months) | (22.5, 7 months) | |
| P | Q | S |
| ↓ | ↓ | ↓ |
- Ratio $(50 \times 4 + 25 \times 8) : (45 \times 5 + 22.5 \times 7) : (70 \times 7)$
of profits
400 : 382.5 : 490 ($\times 0.4$)
160 : 153 : 196
Profit at the end of 12 months = Rs 1272.5

$$S's \text{ share} = \frac{196}{160 + 153 + 196} \times 1272.5$$

$$= \frac{196}{509} \times 1272.5 = \frac{196 \times 1272.5}{509} = \text{Rs. } 490$$

47. (c); A invests Rs. 12000 for 4 months and Rs. 17000 for 8 months B invests Rs. 16000 for 4 months and Rs. 10000 for 8 months. C invests Rs. 20000 for 6 months.

$$A \rightarrow 12000 \times 4 + 17000 \times 8 = 1,84,000$$

$$B \rightarrow 16000 \times 4 + 10000 \times 8 = 1,44,000$$

$$C \rightarrow 20000 \times 6 = 1,20,000$$

$$\therefore \text{Ratio of profits} = 23 : 18 : 15$$

$$\text{Total profit} = \text{Rs. } 30,100$$

$$\text{As share} = \frac{23}{(23 + 18 + 15)} \times 30,100 = \text{Rs. } 12362.5$$

48. (a);
- | | | | |
|----------------------|---|---|---|
| | P | Q | R |
| Ratio of investments | 8 | 6 | 5 |
| Ratio of profits | 1 | 3 | 5 |
- Check option (a) $\rightarrow$ Ratio of time periods
 $\rightarrow 1 : 4 : 8$
Ratio of profits = Ratio of products of corresponding values of investments and time periods
 $8 \times 1 : 6 \times 4 : 5 \times 8$
 $= 8 : 24 : 40$
 $= 1 : 3 : 5 = \text{Ratio of profits.}$

49. (d); Total time = 36 months

Let Aditya's time of contribution = t_A

Manish's time = t_M

$300 \times t_A : 500 \times t_M = \text{Ratio of their profit shares.}$

Total profit = Rs 1020

Aditya's share = 495

Manish's share = $1020 - 495 = 525$

Ratio of profits = $495 : 525 = 33 : 35 \dots (ii)$

From (1) and (2)

$$\frac{300 \times t_A}{500 \times t_M} = \frac{33}{35} \Rightarrow \frac{t_A}{t_M} = \frac{33 \times 500}{35 \times 300} = \frac{11}{7}$$

Time period for which Aditya invested

$$= \frac{11}{18} \times 36 = 22 \text{ months}$$

50. (a);

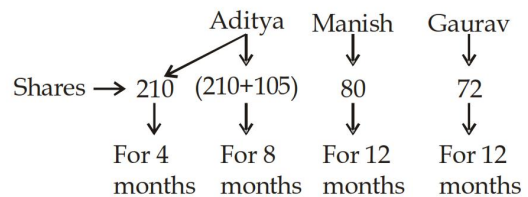
	Aditya	Manish	Gaurav
Ratio of shares	$\frac{7}{2}$	$\frac{4}{3}$	$\frac{6}{5}$

$$\frac{7}{2} \times 30 : \frac{4}{3} \times 30 : \frac{6}{5} \times 30$$

$$(105 : 40 : 36) \times 2$$

$$210 : 80 : 72$$

Let their shares be $\rightarrow 210, 80, 72$



Ratio of $\rightarrow$	Aditya	Manish	Gaurav
profit shares	↓	↓	↓
	$(210 \times 4 + 315 \times 8)$	(80×12)	(72×12)

Ratio of profit $\rightarrow$	↓	↓	↓
	$(840 + 2520)$	960	864
	↓	↓	↓
	3360	960	864

Profit at the end of one year = Rs. 43200

$$\text{Manish's share} = \frac{960}{5184} \times 43200 = \text{Rs. } 8000$$

Difficult

1. (c); Alloy 1 Alloy 2

$$\begin{array}{ccc} \frac{1}{3} & & \frac{2}{5} \\ & \searrow \quad \swarrow & \\ & \frac{3}{8} & \\ & \swarrow \quad \searrow & \\ \frac{2}{5} - \frac{3}{8} = \frac{1}{40} & & \frac{3}{8} - \frac{1}{3} = \frac{1}{24} \end{array}$$

$$\text{Required Ratio} = \frac{1}{40} : \frac{1}{24} = 24 : 40 = 3 : 5$$

3 and 5 parts of the two alloys must be taken.

2. (d); Quantity of gold and silver in first alloy,
Gold = 8 kg, silver = 2kg
In second alloy
Gold = 4 kg, silver = 12 kg
Total amount of gold in new alloy = 8 + 4 = pure gold added
Total amount of silver in new alloy = 12 + 2 = 14 kg
Ratio given = 3 : 2
 $2x = 14 \text{ kg} \Rightarrow x = 7 \text{ kg}$
Total weight of new alloy = $5 \times x = 5 \times 7 = 35 \text{ kg}$

3. (b); $A = \frac{B+C}{2}, \frac{B+C}{A} = \frac{2}{1} \Rightarrow \frac{A+C}{4} = B, \frac{A+C}{B} = \frac{4}{1}$

$$\frac{B+C}{A} = \frac{2 \times 5}{1 \times 5} = \frac{10}{5} \Rightarrow \frac{A+C}{B} = \frac{12}{3}$$

$$B = 3, A = 5, C = 7 \Rightarrow A : B : C = 5 : 3 : 7$$

$$(A + B)'s \text{ share} = \frac{8}{15} \times 5625 = \text{Rs. } 3000$$

4. (d); Average income = $\frac{8 \times 2310 + 4 \times 1800 - 1680}{12}$

$$= \frac{24000}{12} = 2000$$

$$\text{Avg. Expenditure} = \frac{8 \times 2310 + 4 \times 1800}{12}$$

$$= \frac{25680}{12} = 2140$$

$$\frac{\text{Average Income}}{\text{Average Expenditure}} = \frac{2000}{2140} = \frac{100}{107} = 100 : 107$$

5. (a); Given, $\frac{\text{Expense}}{\text{Saving}} = \frac{5}{3}$

Let Income = 8 units

$$\text{Increased income} = 8 \times \frac{5}{4} = 10 \text{ units}$$

Expense is increased by 60%

$$\text{Increased expense} = 5 \times \frac{8}{5} = 8 \text{ unit}$$

New saving = $10 - 8 = 2$ units

Difference between saving = $3 - 2 = 1$ unit

1 unit = Rs. 3500

10 units = Rs. 35000

6. (a); Let the numbers of Rs. 1, 50 paise & 25 paise coins are $12x, 10x$ and $7x$.

According to question

$$12x + \frac{10x}{2} + 7\left(\frac{x}{4}\right) = 75 \Rightarrow 12x + 5x + \frac{7x}{4} = 75$$

$$17x + \frac{7x}{4} = 75 \Rightarrow \frac{75x}{4} = 75, x = 4$$

Number of 25 paise coins = $7 \times 4 = 28$ coins

7. (a); Given, 2 Horses = 5 Dogs

6 Dogs = 8 Oxens

10 Oxens = 50 Sheep

14 Sheep = 9 Goats

$$\text{Cost of one horse} = \frac{9 \times 50 \times 8 \times 5}{14 \times 10 \times 6 \times 2} \times 700 = \text{Rs. } 7500$$

8. (c); Let their share after decrease in money = $9x, 13x$ and $8x$

$$(9x + 26) + (13x + 28) + (8x + 32) = 2186$$

$$30x = 2186 - 26 - 28 - 32 \Rightarrow x = 70$$

Amount given to A = $9 \times 70 + 26$

$$= 630 + 26 = \text{Rs. } 656$$

9. (a); $\frac{\text{Sita's Profit}}{\text{Gita's Profit}} = \frac{3}{2}$

Let Gita invested capital for x months and Sita invested her capital for 12 months.

$$\frac{3}{2} = \frac{12}{x} \Rightarrow x = \frac{12}{3} \times 2 = 8 \Rightarrow x = 8 \text{ months}$$

10. (a); Last year, ratio of cost of Maruti and Figo, $\frac{LM}{LF} = \frac{3}{4}$

Present year & last year ratio of maruti, $\frac{PM}{LM} = \frac{5}{4}$

$$PM = \frac{5}{4}LM, PM = \frac{5}{4} \times 3 = \frac{15}{4} \Rightarrow \frac{PF}{LF} = \frac{3}{2}$$

$$PF = \frac{3}{2}LF \Rightarrow PF = \frac{3}{2} \times 4 = 6$$

$$PF : PM = 6 : \frac{15}{4} = 24 : 15 = 8 : 5$$

$$PF = \frac{8}{13} \times 7.8 = 4.8 \text{ lacks}$$

$$\frac{LF}{PF} = \frac{2}{3}, LF = \frac{2}{3} \times 4.8 = 3.2 \text{ lacs}$$

11. (b); Ratio of Ram's salary for last and present year = 2 : 3

Ratio of Shyam's salary for last and present year = 4 : 5

Ratio of their salary during last year = 3 : 5

Now, we change the first two ratio

$$2 : 3 = \frac{2 \times 3}{2} : \frac{3 \times 3}{2} = 3 : \frac{9}{2}$$

$$\text{and } 4 : 5 = 4 \left(\frac{5}{4} \right) : 5 \left(\frac{5}{4} \right) = 5 : \frac{25}{4}$$

Now the ratio of salary of present year

$$= \frac{9}{2} : \frac{25}{4} = 18 : 25$$

$$\text{The Present salary of Ram} = \frac{18}{18+25} \times 8600$$

$$= \text{Rs. } 3600$$

12. (d); Let the candle height be x
Let time required = T hour

$$\text{In one hour first candle lighted} = \frac{x}{8}$$

$$\text{In T hour, first candle lighted} = \frac{T}{8}x$$

$$\text{After T hour, remaining candle} = x \left(1 - \frac{T}{8} \right)$$

$$\text{Similarly for second candle} = x \left(1 - \frac{T}{6} \right)$$

$$\frac{x \left(1 - \frac{T}{8} \right)}{x \left(1 - \frac{T}{6} \right)} = \frac{2}{1}, \left(\frac{1 - \frac{T}{8}}{1 - \frac{T}{6}} \right) = \frac{2}{1}$$

$$1 - \frac{T}{8} = 2 - \frac{T}{3}, \frac{T}{3} - \frac{T}{8} = 1$$

$$T = \frac{24}{5} = 4 \text{ h } 48 \text{ min}$$

13. (a); Income of Ram, Shyam and Mohan be x, y and z respectively.

$$\text{Ram's Saving} = 75\% \text{ of } x = \frac{3}{4}x$$

$$\text{Shyam's Saving} = 80\% \text{ of } y = \frac{4}{5}y$$

$$\text{Mohan Saving} = 50\% \text{ of } z = \frac{1}{2}z$$

Let their saving be 9k, 8k and 4k then

$$\frac{3}{4}x = 9k, x = 12k, \Rightarrow \frac{4}{5}y = 8k, y = 10k$$

$$\frac{1}{2}z = 4k, z = 8k$$

According to question

$$x + y + z = 450 \Rightarrow 12k + 10k + 8k = 450$$

$$k = \frac{450}{30} = 15$$

$$\text{Ram's income} = 12 \times 15 = \text{Rs. } 180$$

14. (a); After taking out 100 litres of milk and adding same amount of water.

$$\text{Ratio of milk and water} = \frac{900}{100} = 9 : 1$$

After taking out 200 litres of mixtures, then the

$$\text{Quantity of milk} = 800 \left(\frac{9}{9+1} \right) = 720 \text{ litres}$$

$$\text{The new ratio} = \frac{720}{280} = \frac{18}{7} = 18 : 7$$

After taking out 400 litres of mixture the rest amount of mixture will be 600.

$$\text{Quantity of pure milk} = 600 \left(\frac{18}{25} \right) = 24 \times 18 = 432$$

litre

The amount of pure milk will be 432 litres.

15. (b); Let the land and water part on earth be x and 2x.
Total area = 3x

$$\text{Total area of northern hemisphere} = \frac{3}{2}x$$

Land part of northern hemisphere

$$= \frac{2}{2+3} \times \frac{3x}{2} = \frac{3x}{5}$$

$$\text{Land part of southern hemisphere} = x - \frac{3x}{5} = \frac{2x}{5}$$

Water part of northern hemisphere

$$= \frac{3}{2+3} \times \frac{3x}{2} = \frac{9x}{10}$$

Water part of southern hemisphere

$$= 2x - \frac{9x}{10} = \frac{11x}{10}$$

$$\text{Required ratio} = \frac{\frac{2x}{5}}{\frac{11x}{10}} = 2 : \frac{11}{2} = 4 : 11$$

16. (a); Income of A = Rs. 2400

$$\text{Income of B} = \frac{7}{3} \times 2400 = \text{Rs. } 5600$$

$$\text{Income of C} = \frac{4}{3} \times 2400 = \text{Rs. } 3200$$

$$\text{Expenditure of A} = 2400 - 300 = \text{Rs. } 2100$$

$$\text{Expenditure of B} = \frac{3}{4} \times 2100 = \text{Rs. } 1575$$

$$\text{Expenditure of C} = \frac{5}{4} \times 2100 = \text{Rs. } 2625$$

$$\text{Saving of B} = \text{Rs. } (5600 - 1575) = \text{Rs. } 4025$$

$$\text{Saving of C} = \text{Rs. } (3200 - 2625) = \text{Rs. } 575$$

17. (b); Let the quantity of wine and water in drum are $3x$ and x respectively.

and let y litre of mixture is taken out and y litres water is mixed.

In y litre of mixture,

$$\text{wine} = \frac{3}{4}y \text{ litre, water} = \frac{1}{4}y \text{ litre}$$

$$\text{amount of wine in new mixture} = 3x - \frac{3y}{4} \text{ litre}$$

$$\text{and amount of water} = x - \frac{y}{4} + y = x + \frac{3y}{4}$$

According to question

$$3x - \frac{3y}{4} = x + \frac{3y}{4} \Rightarrow 2x = \frac{3y}{2}, y = \frac{4x}{3}$$

Which is $\frac{1}{3}$ of the total mixture.

18. (a);

Petrol	Kerosene	Total mixture
99	a	99 + a
99	a - 198	a - 99

$$\frac{99}{a - 99} \times 100 - \frac{99}{a + 99} \times 100 = 13.33$$

$$\frac{99}{a - 99} \times 100 - \frac{99}{a + 99} \times 100 = 13.33$$

$$9900 \left(\frac{198}{a^2 - 99^2} \right) = 13.33$$

$$a^2 - 99^2 = 99^2 \times 15 \Rightarrow a^2 = 99^2 \times 16 \Rightarrow a = 396 \text{ litres}$$

$$\text{Actual concentration} = \frac{99}{(396 + 99)} \times 100 = 20\%$$

19. (a); Here, $B = \frac{5}{6}A$, $C = D = \frac{9}{10}B$

$$B = \frac{2}{3}E \text{ and } E - A = 3 \Rightarrow B = \frac{5}{6}A = \frac{2}{3}E$$

$$\frac{A}{E} = \frac{4}{5} \text{ or } E = \frac{5}{4}A \Rightarrow E - A = \frac{5}{4}A - A$$

$$3 = \frac{1}{4}A, A = 12$$

$$E = 15 \text{ and } B = 10, C = D = 9 \text{ and } B = 10$$

$$C = D = 9 \text{ and } F = 11, A : F = 12 : 11$$

20. (d); In first Vessel $\frac{W_1}{M_1} = \frac{1}{3}$

$$\text{In second Vessel } \frac{W_2}{M_2} = \frac{2}{3}$$

$$\text{In third Vessel } \frac{W_3}{M_3} = \frac{2}{5}$$

$$\text{Proportion of water} = \frac{1}{4} : \frac{2}{5} : \frac{2}{7}$$

$$\frac{1}{4} \times \frac{5 \times 7}{5 \times 7} : \frac{2}{5} \times \frac{4 \times 7}{4 \times 7} : \frac{2}{7} \times \frac{4 \times 5}{4 \times 5}$$

$$\frac{35}{140} : \frac{56}{140} : \frac{40}{140}$$

Now, since all these mixtures are mixed in ratio of 2 : 3 : 5.

$$\text{Therefore new ratio} = \frac{70}{280} : \frac{168}{420} : \frac{200}{700}$$

$$\text{Amount of water} = 70 + 168 + 200 = 438$$

$$\text{Amount of milk} = (280 + 420 + 700) - 438 = 962$$

$$\text{Ratio of milk to water} = \frac{962}{438} = \frac{481}{219} = 481 : 219$$

Previous Year (Memory Based)

1. (b); $A : B = 5 : 7$

$$B : C = 6 : 11$$

So, we have to quantity B which to be equal in both case.

$$\text{So, } A : B = [5 : 7] \times 6$$

$$B : C = [6 : 11] \times 7$$

$$\text{So, } A : B : C = 30 : 42 : 77$$

Shortcut:

$$\begin{array}{ccc} A & B & C \\ 5 & \downarrow 7 & \searrow 11 \\ & 6 & \end{array}$$

$$\text{So, } A : B : C = 5 \times 6 : 7 \times 6 : 7 \times 11 \\ = 30 : 42 : 77$$

2. (d); Let $A = 2x$, $B = 3x$ and $C = 4x$

$$\text{So, } \frac{A}{B} = \frac{2x}{3x}, \frac{B}{C} = \frac{3x}{4x} \text{ and } \frac{C}{A} = \frac{4x}{2x}$$

$$\Rightarrow \frac{A}{B} : \frac{B}{C} : \frac{C}{A} = \frac{2}{3} : \frac{3}{4} : \frac{4}{2} = 8 : 9 : 24$$

3. (b); $\frac{4^{3.5}}{2^5} = \frac{2^{(2 \times 3.5)}}{2^5} = \frac{2^7}{2^5} = \frac{4}{1}$

4. (a); $\frac{x}{y} = \frac{2}{1}, \frac{x^2}{y^2} = \frac{4}{1} \Rightarrow \frac{x^2 + y^2}{x^2 - y^2} = \frac{4 + 1}{4 - 1}$

$$\frac{x^2 - y^2}{x^2 + y^2} = \frac{3}{5}$$

5. (c); $19(4x^2 - 3y^2) = 12(2x^2 + 5y^2)$
 $52x^2 = 117y^2 \Rightarrow 4x^2 = 9y^2$

$$\frac{x}{y} = \frac{3}{2}$$

6. (a); Let $(a + b) = 6k$, $(b + c) = 7k$ and $(c + a) = 8k$
 then, $2(a + b + c) = 21k$

$$2 \times 14 = 21k \Rightarrow k = \frac{4}{3}$$

$$\text{So, } (a + b) = 6 \times \frac{4}{3} = 8$$

$$c = (a + b + c) - (a + b) = 14 - 8 = 6$$

7. (d); Given ratio = $\frac{1}{2} : \frac{2}{3} : \frac{3}{4} = 6 : 8 : 9$

$$\text{So, 1st part is } = \frac{6}{23} \times 782 = 204$$

8. (d); Let the original salaries of Ravi and Sumit be $2x$ and $3x$.

$$\text{Then, } \frac{2x + 4000}{3x + 4000} = \frac{40}{57}$$

$$3x = 34000$$

$$\text{Sumit's present salary} = (3x + 4000) = 38000$$

9. (b); $A = \frac{2}{3}B$, $B = \frac{1}{4}C$

$$\text{So, } A : B : C = 2 : 3 : 12$$

$$A's \text{ share} = \frac{2}{17} \times 510 = 60$$

$$B's \text{ share} = \frac{3}{17} \times 510 = 90$$

$$C's \text{ share} = \frac{12}{17} \times 510 = 360$$

10. (c); Let Boys = $3x$ and girls = $2x$

No. of students who do not get scholarship

$$= (80\% \text{ of } 3x) + (75\% \text{ of } 2x) = \frac{39x}{10}$$

$$\text{Req. percentage} = \left(\frac{39x}{10} \times \frac{1}{5x} \times 100 \right) \% = 78\%$$

11. (b); $10\% \text{ of } B = \frac{1}{4}G$

$$\frac{B}{G} = \frac{5}{2}$$

12. (b); Let fixed amount = x

Cast of each unit = y

$$540y + x = 1800 \quad \dots(i)$$

$$620y + x = 2040 \quad \dots(ii)$$

from (i) and (ii)

$$x = 180, y = 3$$

$$\text{So, total charges for consuming 500 units} \\ = (180 + 500 \times 3) = 1680$$

13. (b); Let C gets Rs. x

and B gets Rs. $(x + 8)$

A gets Rs. $(x + 15)$

$$\text{So, Sum of } A + B + C = 53$$

$$x + x + 8 + x + 15 = 53$$

$$x = 10$$

$$\text{So, } A : B : C = 25 : 18 : 10$$

14. (c); Let the no. of 25p, 10p and 5p coins be x, 2x and 3x respectively.

So, Sum of their values

$$= \left(\frac{25x}{100} + \frac{10 \times 2x}{100} + \frac{5 \times 3x}{100} \right) = \frac{60x}{100}$$

$$\frac{60x}{100} = 30$$

So, x = 50

So, the no. of 5p coins = 3 × 50 = 150

Shortcut:

$$\begin{array}{ccccccc} & 25p & & 10P & & 5P & \\ \text{No. of coins} & 1 & : & 2 & : & 3 & = \frac{300}{6} \end{array}$$

So, No. of 5p coins = 50 × 3 = 150 coins

15. (b); From the question

$$50 \times 2K + 100 \times 3K + 500 \times 4K + 1000 \times K = 34000$$

$$K = 10$$

So, No. of 500 notes = 4 × 10 = 40

16. (a); From the Question $\frac{C}{A+B} = \frac{3}{4}$, $\frac{B}{A+C} = \frac{12}{23}$

$$\frac{C}{A+B} = \frac{3}{4}$$

So, the sum of both term of the ratio

$$C + A + B = 3 + 4 = 7$$

$$\text{So, runs scored by } C = \frac{3}{7} \times 1750 = 750 \text{ runs}$$

Similarly,

$$\text{and runs scored by } B = \frac{12}{12+23} \times 1750 = 600 \text{ runs}$$

$$\text{and runs scored by } A = 1750 - 750 - 600 = 400 \text{ run}$$

17. (c); No. of employees in class - 1 = $\frac{5}{17} \times 68 = 20$

$$\text{No. of employees in class - 2} = \frac{10}{17} \times 68 = 40$$

$$\text{No. of employees in class - 3} = \frac{2}{17} \times 68 = 8$$

So, Salary of each class employee be 2x, 3x and 5x.

$$\text{So, } 20 \times 2x + 40 \times 3x + 8 \times 5x = 400000$$

$$x = 2000$$

So, salary of employee of class - 3 is = 10000

18. (a); According to the Question

$$20 \times 2x + 40 \times 3x + 8 \times 5x = 400$$

$$x = 2$$

So, he spent $20 \times 2 \times 2$, $40 \times 3 \times 2$, $8 \times 5 \times 2$ on those three vehicles respectively.

So, the spent on car = 240

19. (a); Price $\propto$ (weight)²

$$P \propto w^2, \quad P = k \cdot w^2$$

$$P = K \cdot (10x)^2$$

...(i)

Now, total price of broken pieces of diamond

$$P_n = K[x^2 + (2x)^2 + (3x)^2 + (4x)^2] = 30 K \cdot x^2 \quad \dots(ii)$$

A.T.Q.

$$P - P_n = 2,10,000$$

$$K \cdot 100x^2 - K \cdot 30 \cdot x^2 = 2,10,000$$

$$70 \cdot K \cdot x^2 = 2,10,000$$

$$\text{Price of unbroken diamond} = 100K \cdot x^2$$

$$= 3,00,000$$

20. (a); $\frac{a+b}{5} = \frac{b+c}{6} = \frac{c+a}{7} = \frac{a+b+b+c+c+a}{5+6+7}$

$$= \frac{2(a+b+c)}{18}$$

$$a+b = 5x,$$

$$a+b+c = 9x$$

$$b+c = 6x$$

$$\text{and } c+a = 7x$$

$$\text{So, } a : b : c = 3 : 2 : 4$$

$$\text{Now, } a+b+c = 18$$

$$\text{So, the value of } x = \frac{18}{9} = 2$$

$$\text{So, } b = 2 \times 2 = 4$$

21. (a); $2a = 3b$

$$\frac{a}{b} = \frac{3}{2}, \quad \frac{b}{c} = \frac{4}{3}$$

$$\text{So, } a : b : c = 6 : 4 : 3$$

22. (a); The ratio of

$$1 \text{ rupees} : 50 \text{ paise} : 25 \text{ paise} = \frac{5}{6} \times \frac{3}{5} : \frac{3}{5} : 1$$

$$= 5 : 6 : 10$$

From the Question

$$5K + \frac{50}{100} \times 6K + \frac{25}{100} \times 10K = 262.50$$

$$K = 25$$

So, the no. of 25 paise coin is = 250

23. (a); $\frac{\text{male amount}}{\text{female amount}} = \frac{4}{3}$

$$\text{and } \frac{\text{female amount}}{\text{children amount}} = \frac{6}{5}$$

So, ratio of male amount : female amount : children = 8 : 6 : 5

from the question,

$$10 \times 8K + 15 \times 6K + 5 \times 5K = 11700$$

$$K = 60$$

So, each child get = 300

24. (a); From the question

$$\frac{5x - 2500}{4x - 2500} = \frac{10}{7}$$

$$x = 1500$$

$$\text{Shyam's income} = 4 \times 1500 = 6000$$

25. (a); 1 cows = 3 calves

The ratio of the cattles of A, B and C.

$$= 4 \times 3 + 3 : 5 \times 3 + 4 : 6 \times 3 + 5$$

$$= 15 : 19 : 23$$

$$\text{Ratio of rents} = 15 \times 5 : 19 \times 3 : 23 \times 2$$

$$= 75 : 57 : 46$$

$$\text{Rent of B} = \frac{2670 \times 57}{75 + 57 + 46} = 855$$

The difference between rent of A and C

$$= \frac{75 - 46}{75 + 57 + 46} \times 2670 = 435$$

26. (a); The ratio of capitals of OM, Jai, Jagdish
 $= 5000 \times 12 : 3000 \times 4 + 4000 \times 8 : 7000 \times 8 + 3000 \times 4$
 $= 15 : 11 : 17$

$$\text{So, Jagdish's share} = \frac{17}{43} \times 34400 = 13600$$

27. (b); Let Ram, Krishna and Ganesh invested $5x$, $4x$ and $3x$.

$$\text{Ram's Capital} = 5x \times 3 + 3x \times 6 = 33x$$

$$\text{Krishna's Capital} = 4x \times 4 + 3x \times 5 = 31x$$

$$\text{Ganesh's Capital} = 3x \times 4 + 4x \times 5 = 32x$$

So, clearly shows Max. profit gained by Ram.

$$\text{Ram's profit} = \frac{33}{96} \times 19200 = 6600$$

28. (c); The amount given to B as management fees = 12% of 7500 = 900
the rest amount of profit = 7500 - 900 = 6600

$$\text{Ratio of capitals of A and B} = \frac{12000}{14400} = \frac{5}{6}$$

$$\text{A's share} = \frac{5}{5+6} \times 6600 = 3000$$

$$\text{B's share} = 7500 - 3000 = 4500$$

So, Difference between A and B share is = 1500

29. (a); Let the capital of Laxmi and Saraswati be $7x$ and $5x$.

Let, Saraswati's capital was invested for y months

So, the ratio of their capital = $7x \times 20 : 5x \times y$

$$\frac{140x}{5xy} = \frac{4}{3}, y = 21 \text{ months}$$

$$30. (c); \frac{p}{q} = \frac{r}{s} = \frac{t}{u} = \frac{2}{3} \Rightarrow \frac{mp}{mq} = \frac{nr}{ns} = \frac{ot}{ou} = \frac{2}{3}$$

$$\Rightarrow \frac{mp + nr + ot}{mq + ns + ou} = \frac{2+2+2}{3+3+3} = \frac{2}{3}$$

$$31. (a); \frac{\sqrt{2}}{(1+\sqrt{3})} = \frac{\sqrt{6}}{x}$$

$$x = \sqrt{3}(1+\sqrt{3})$$

$$x = 3 + \sqrt{3}$$

32. (d); Ratio of Vinay and Aditya for one month
 $= (50000 \times 12) + (80000 \times 24) : (70000 \times 24)$
 $= 600000 + 1920000 : 1680000 = 3 : 2$
Hence, share of Aditya in the profit earned from

$$\text{the business} = \frac{87500}{3+2} \times 2 = 35000$$

33. (b); Let the score of Ajay = x

$$\text{Rahul} = x - 15$$

$$\text{Manish} = x - 25$$

As per the question

$$x = 63 + 30 = 93$$

$$\text{score of Ajay} = 93$$

$$\text{Rahul} = 93 - 15 = 78$$

$$\text{Manish} = 93 - 25 = 68$$

Total marks of Rahul, Manish and Suresh

$$= 3 \times 63 = 189$$

$$\text{Suresh} = 189 - (78 + 68) = 43$$

$$\text{Manish} + \text{Suresh} = 68 + 43 = 111$$

34. (d); Let A's amount be $5x$

$$\text{B's amount be } 6x$$

Again let B invested the capital for y months.

$$\frac{5x \times 8}{6x \times y} = \frac{5}{9} \Rightarrow \frac{40}{6y} = \frac{5}{9}$$

$$y = \frac{9 \times 40}{6 \times 5} = 12 \text{ months}$$

35. (d); Ratio of profit among A, B and C
 $= (42000 \times 4 + 30000 \times 6) : (30000 \times 4 + 24000 \times 6)$
 $: (28000 \times 4 + 20000 \times 6)$
 $= 348 : 264 : 232$

$$\text{Hence C's share} = \frac{46420}{844} \times 232 = 12760$$

36. (c); Difference of amount received by R and Q is $(7 - 5) = 2$

Total amount received P and Q = $(3 + 5) = 8$

Then, 2 corresponds to 4000 implies that 8

corresponds to $\frac{4000}{2} \times 8 = 16000$

37. (d); Cat Dog

5 ← 4
4 → 3

15 : 16

38. (a); Total of max. marks of all subjects = 105×5
= 525

$$80\% \text{ of } 525 = \frac{525 \times 80}{100} = 420$$

Obtained marks of 4 subjects = 360

So, marks obtained in science = $420 - 360 = 60$

39. (c); There are 70 males out of 120 applicants, there must be 50 females.

Thus, no. of males having a driver's licence all 50 females must having a driver's licence will be $80 - 50 = 30$. The max. possible no. of males having a driving license is 70.

So, ratio between the min. and max. is 30 : 70 or 3 : 7.

40. (b); Given ratio of employees = 9 : 13 : 18

A.T.Q.

$$18x = 54$$

$$x = 3$$

No. of employees of type A = $9 \times 3 = 27$

Similarly, wages of every employee of type A

$$= \frac{10}{7} \times 1400 = 2000$$

$$\text{Req. wages} = 27 \times 2000 = 54000$$